Nonintrusive Cross-Motor Detection and Severity Assessment of False-Data-Injection Attacks in Heterogeneous Inverter-Fed Drives

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Blue underlined text is added or revised.
~~Red struck-out text is removed.~~

Unmarked text is unchanged. Tables and figures are paired as complete ORIGINAL and REVISED blocks (left/right or top/bottom). Cross-references follow the revised numbering. Formatting-only changes and commented-out source text are not marked word by word.

***Abstract*—**

This paper presents an experimentally validated method for detecting false-data-injection attacks (FDIA) and estimating attack severity in heterogeneous motor-drive systems using measurements acquired either at the motor terminals or at the point-of-common-coupling. The method extracts multi-scale features from variable-size windows and uses an InceptionTime model for binary attack detection and four-level severity classification. Motor-terminal currents and PCC voltage/current are evaluated separately using three partitions with no shared recordings or episodes between training, validation, and test sets. Experiments on a two-branch platform containing one alternating current induction motor (ACIM) drive and one permanent magnet synchronous motor (PMSM) drive show that motor-side sensing yields the strongest class separability, while point-of-common-coupling (PCC) sensing remains attractive for nonintrusive deployment. In the mixed-domain setting, binary detection reaches ~~98.89%~~ 87.41% mean accuracy with motor-side data and ~~89.44%~~ 80.78% mean accuracy with PCC data. For four-class severity assessment, the motor-side stream achieves ~~98.68%~~ 87.64% mean accuracy, while the PCC stream achieves ~~84.17%~~ 61.00% mean accuracy. Among the evaluated deep-learning and classical machine-learning baselines, the proposed method achieved the ~~best PCC-side performance~~ highest mean PCC balanced accuracy for both binary detection and four-level severity assessment. ~~For the PCC-side tasks, the proposed method outperformed the evaluated baselines, achieving 89.44% accuracy for binary detection and 84.17% accuracy for four-level severity classification, compared with the best baseline accuracies of 86.45% and 83.42%, respectively.~~ In cross-motor experiments, few-shot fine-tuning improved transfer performance, especially in the PMSM-to-ACIM direction, where a nominal 1% target-episode budget raised PCC balanced accuracy from 56.00% to 77.43%. These results indicate that PCC-based FDIA monitoring is promising on the tested ACIM/PMSM platform.

***Index Terms*—False-data-injection attack, point of common coupling, inverter-fed motor drives, heterogeneous motor drives, nonintrusive monitoring, cross-motor adaptation, severity assessment, cyber-physical security**

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I. Introduction

FDIA detection and severity assessment are important in networked motor-drive systems, where manipulated sensor feedback can degrade control performance and cause unsafe operation [1]. Increasing drive connectivity makes timely detection of sensor-oriented cyber attacks essential for resilient operation [2], [3]. Current signals are attractive for FDIA monitoring because they are easy to acquire, strongly coupled to electromechanical dynamics, and have been shown to reveal sensor-integrity attacks in motor drives [4], [5]. However, most existing studies emphasize terminal-side sensing, operate on a single machine domain, or assume sufficient labeled data for supervised training. These assumptions limit practicality in heterogeneous deployments where different motor types, sensing locations, and data availability constraints all affect performance. Motivated by these limitations, this work experimentally evaluates FDIA detection in a heterogeneous motor-drive setting involving both ACIM and PMSM domains. The study evaluates binary detection (`fdia2`), four-level severity assessment (`severity4`), cross-domain transfer, and deployment feasibility. The framework combines low-frequency (LF), high-frequency (HF), and time-domain (TD) ~~current~~ electrical features with an InceptionTime classifier, and compares motor-terminal and PCC sensing. A three-class classifier (`fdia3`) additionally distinguishes normal operation, an ACIM attack, and a PMSM attack.

A. Challenges at PCC in Heterogeneous Multi-Motor Systems

From a deployment perspective, motor-terminal measurements provide cleaner machine-specific signatures, whereas PCC measurements are more scalable and non-intrusive. PCC sensing is attractive because it avoids direct motor-terminal instrumentation and is compatible with shared electrical infrastructure [6]. However, PCC currents are affected by converter harmonics, network interactions, and load aggregation. Consequently, PCC attack signatures are weaker and less separable than motor-terminal signatures [7]. The problem becomes more difficult in heterogeneous multi-motor settings. ACIM and PMSM systems exhibit different current characteristics even under the same attack framework, which introduces a substantial domain shift between source and target deployments. Consequently, a detector trained on one motor family may generalize poorly to the other unless explicit adaptation is used. Therefore, this study evaluates in-domain

performance, source-only transfer, few-shot adaptation, and transfer-direction asymmetry.

B. Contributions

The main contributions of this work are as follows:

- ~~**Developedand validated a**~~ **Experimental PCC-based** ~~**FDIA monitoring framework for heterogeneous inverter-fed drives, showing that PCC sensing supports useful binary detection and coarse four-level severity assessment without direct motor-terminal instrumentation**~~**FDIA monitoring:** A multiscale feature and InceptionTime framework is validated using 540 attack episodes across three speeds and three load settings for each motor type. Complete recordings and their episodes are separated between training, validation, and testing, enabling evaluation on unseen recordings without temporal leakage between these sets.
- ~~**Constructed**~~**Detection and severity assessment at a** ~~**multi-scale LF/HF/TD current-feature representation and showed that it improves PCC-side separability under harmonic distortion and signal aggregation, while also enabling stronger motor-side performance**~~**shared sensing point:** Separate motor-terminal and PCC models quantify the sensing tradeoff. PCC monitoring detects attacks without dedicated motor-terminal instrumentation, while severity-conditioned attack recall identifies attacks even when their exact bias level is misclassified.
- ~~**Provided**~~**Limited-label adaptation and reduced-feature inference:** ~~a~~ Bidirectional cross-motor ~~evaluation across ACIM and PMSM domains, including source-only transfer, few-shot target-domain adaptation, and latency/pruning analysis, showing both the limits of direct transfer and the practical gains from lightweight adaptation~~experiments show that a nominal 1% target-episode budget improves PMSM-to-ACIM PCC balanced accuracy by 21.43 percentage points. Validation-based feature pruning preserves PCC detection performance with 18 of 74 features and an 85.18% processing speedup.

The remainder of this paper is organized as follows: Section II reviews related literature. Section III describes the physical system and dataset construction, while Section IV details the proposed FDIA detection framework and feature pipeline. Experimental results and cross-domain adaptations are reported in Section V. Section VI analyzes real-time deployment and latency, followed by a discussion of tradeoffs in Section VII. Section VIII concludes the paper.

II. Related Work

FDIA detection has been studied extensively in smart grids and broader cyber-physical systems (CPS), but the assumptions in that literature do not transfer cleanly to inverter-fed motor drives. Many grid- and CPS-oriented approaches rely on state estimation, observer-based detection, invariant analysis, adaptive neuro-fuzzy models, or graph-based learning [8], [9], [10], [11], [12], [13]. These studies demonstrate strong FDIA-detection capability at the system level, but they are generally not designed for drive-level current monitoring or for deployment at a shared point of common coupling (PCC).

Motor-drive-specific studies are more closely related to the present work. Prior work has shown that current signals can support effective FDIA detection in electric-drive systems and that transfer learning can improve performance when labeled data are limited [2], [3]. Related industrial-IoT monitoring studies have also considered cyber-detection in machine-monitoring settings [14]. However, these studies mainly rely on motor-terminal sensing, single-motor settings, or closely related operating domains, and therefore provide limited evidence for heterogeneous cross-motor transfer.

The main gap is not FDIA detection in general, which has been widely studied in smart grids and cyber-physical systems, but nonintrusive PCC-based monitoring for heterogeneous inverter-fed motor-drive systems. In particular, the existing literature does not clearly establish: 1) how PCC sensing compares with motor-side sensing for low-severity current-feedback FDIAs, 2) whether PCC measurements support reliable multi-level severity assessment rather than only binary detection, given that attack signatures become weaker and less separable under signal aggregation and harmonic distortion, and 3) how detection performance transfers between heterogeneous ACIM and PMSM domains when labeled target-domain data are limited. This work addresses these three questions experimentally on a shared ACIM/PMSM platform using multi-scale features, in-domain evaluation, source-only cross-domain transfer, and few-shot target-domain adaptation.

III. System and Data

A. Experimental Platform: ACIM/PMSM and PCC Sensing

Fig. 1 illustrates the overall setup, including the shared PCC, the two drive branches, the ACIM and PMSM loads, the sensing locations, and the FDIA injection points in the control/feedback paths. The experimental platform consists of a heterogeneous two-motor industrial setup including one ACIM branch and one PMSM branch. Both branches are supplied from a common upstream electrical interface, enabling measurements at the PCC in addition to motor-side measurements. This setup enables comparison of motor-terminal and PCC sensing under heterogeneous ACIM/PMSM dynamics. ~~A total of~~ The dataset contains 540 ~~unique FDIA cases were collected, including~~ scheduled attack episodes (270 ~~ACIM cases and 270 PMSM cases~~for each motor), formed by repeating 30 bias patterns over nine speed/load combinations per motor. All episodes were acquired on the same two-motor platform. Both motors operated throughout: only the ACIM was attacked in ACIM-labeled recordings, and only the PMSM was attacked in PMSM-labeled recordings. Both motors were attack-free during normal intervals. The experiments cover three speed operating points, namely 900, 1100, and 1300 ~~rpm~~RPM.

Two measurement streams were used throughout this study. The first stream corresponds to motor-side measurements, referred to as *motor* in the remainder of the paper. The second stream corresponds to PCC-side measurements, referred to as *PCC*. The PCC stream is the primary deployment-oriented

sensing location because it avoids dedicated sensors at each machine. ~~The measured signals and channel counts associated with the motor-side and PCC-side streams are summarized in Table II.~~

B. Attack Model, Sensing Assumptions, and Labels

In the testbed, the FDIA is injected into the controller's phase-current feedback path. The PCC and motor-side monitoring sensors are used as trusted measurements to record the physical response of the drive system under attack. The monitoring sensors are isolated from the compromised controller feedback path. Thus, PCC sensing represents the nonintrusive monitoring case, while motor-side sensing serves as a performance reference. Replay, denial-of-service, monitoring-sensor attacks, and malicious command manipulation are outside the scope of this study.

Let $\mathbf{i}_{abc}(t) = [i_a(t), i_b(t), i_c(t)]^\top$ denote the true three-phase current-feedback vector in the controller's per-unit representation, and let $\tilde{\mathbf{i}}_{abc}(t)$ denote the corrupted current-feedback vector received by the motor controller. During an attack interval, the FDIA is modeled as an additive phase-current bias:

$$\tilde{\mathbf{i}}_{abc}(t) = \mathbf{i}_{abc}(t) + \boldsymbol{\alpha}, \qquad t \in [t_{\text{on}}, t_{\text{off}}], \tag{1}$$

Here, $\boldsymbol{\alpha} = [\alpha_a, \alpha_b, \alpha_c]^\top$ denotes the injected per-unit bias vector, with $\boldsymbol{\alpha} = \mathbf{0}$ outside the attack interval. The tested nonzero bias magnitudes range from 0.0005 to 0.0050 pu, corresponding to 0.05%–0.5% current-feedback bias. These low-severity FDIAs are difficult to detect because their signatures can be masked by normal variation, harmonics, and PCC aggregation. The bias is normalized to the nominal controller current-feedback base. In this study, severity is defined by the injected bias magnitude; physical consequences such as shaft-torque deviation are not used as severity labels. Attack severity is measured by the largest absolute injected phase-current bias, $s = \|\boldsymbol{\alpha}\|_\infty$.

For binary FDIA detection, each window is labeled as $y_{\text{fdia2}} = 1$ if $s > 0$ and $y_{\text{fdia2}} = 0$ otherwise. For four-class severity assessment, the label is defined as

$$y_{\text{severity4}} = \begin{cases} 0, & s = 0 \quad \text{(normal)}, \\ 1, & 0.0005 \le s \le 0.0015 \quad \text{(mild)}, \\ 2, & 0.0020 \le s \le 0.0035 \quad \text{(medium)}, \\ 3, & 0.0040 \le s \le 0.0050 \quad \text{(severe)}. \end{cases}$$

$$y_{\text{severity4}} = \begin{cases} 0, & s = 0 \quad \text{(normal)}, \\ 1, & 0.0005 \le s \le 0.0015 \quad \text{(low)}, \\ 2, & 0.0020 \le s \le 0.0035 \quad \text{(medium)}, \\ 3, & 0.0040 \le s \le 0.0050 \quad \text{(high)}. \end{cases} \tag{2}$$

~~corresponding to normal,mild,~~Cases 21–30 have $s = 0.0050$ throughout because phase A remains fixed; all belong to the high-bias class. The resulting attack-window counts are 5,400 low, 7,200 medium, and ~~severe conditions,respectively~~14,400 high. Table I summarizes the swept single-phase, equal two-phase, and fixed-plus-swept two-phase FDIA cases. Across three speeds, three load settings, and two motor types, the dataset contains 540 total attack cases. Each recording ~~consists of~~ contains 30 consecutive 20-s episodes, with 10 s normal operation followed by 10 s FDIA operation. In total, the dataset contains 33,054 normal and 27,000 attack windows.

C. Data Acquisition and Operating Conditions

Electrical measurements were acquired from the testbed using a National Instruments cDAQ-9185 chassis and a custom data-collector interface board. The interface board routed synchronized motor-side and PCC-side analog channels to the acquisition system. Table II summarizes the measured signals and channel counts. Three brake-excitation settings were tested for each machine: 1–3 V for the ACIM branch and 2–4 V for the PMSM branch. Because torque was not directly measured, these brake-excitation voltages are used as load-condition labels. All channels were sampled at 10 kHz and stored in TDMS format. For learning-based analysis, each record was segmented into non-overlapping 0.2 s windows containing 2,000 samples. ~~This non-overlapping segmentation prevents temporal leakagebefore train/validation/test splitting~~To prevent temporal leakage, complete recordings and all their episodes were assigned to separate training, validation, and test sets. Normalization statistics were fitted using training recordings only. The resulting dataset dimensions, input configurations, severity-label distributions, and evaluation protocols are summarized in Table III. Three seeds (1234, 2234, and 3234) changed the recording assignment and model initialization; recordings can recur across repeats.

IV. The Proposed Method

A. LF/HF Feature Extraction Pipeline

To represent both slow electromechanical behavior and faster attack-induced disturbances, a two-scale LF/HF feature pipeline was applied independently to each ~~current~~ measured channel. For every sample, a 2,000-point outer window was first extracted from the raw 10 kHz recording. Within this outer window, two parallel sliding analyses were performed. The LF branch uses a 500-sample window with a stride of 25 samples, while the HF branch uses a 250-sample window with a stride of 29 samples. The LF and HF settings both produce 61 time steps, enabling concatenation into one temporal feature tensor. Fig. 2 illustrates representative attack-onset responses and the corresponding LF/HF windows. In both cases, the raw signal is shown around the attack-transition instant, and representative LF and HF windows are overlaid to illustrate the two-scale extraction procedure. ~~The motor-side signals exhibit stronger and more localized variations, whereas PCC-side responses are more distributed due to their aggregated nature~~Signal amplitudes are in recorded acquisition units.

In total, 37 features were extracted per channel, including 18 LF spectral features, 11 HF spectral/envelope features, and 8 TD features. The LF branch targets slow electromechanical changes, fundamental-frequency behavior, harmonic

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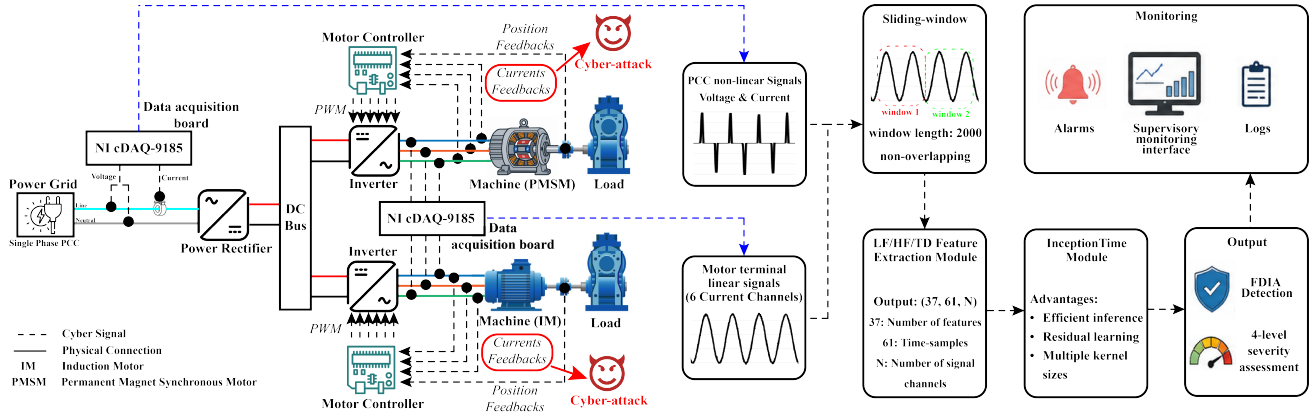


Figure 1: Experimental platform used to generate current-feedback FDIAs in a heterogeneous inverter-fed motor-drive testbed.

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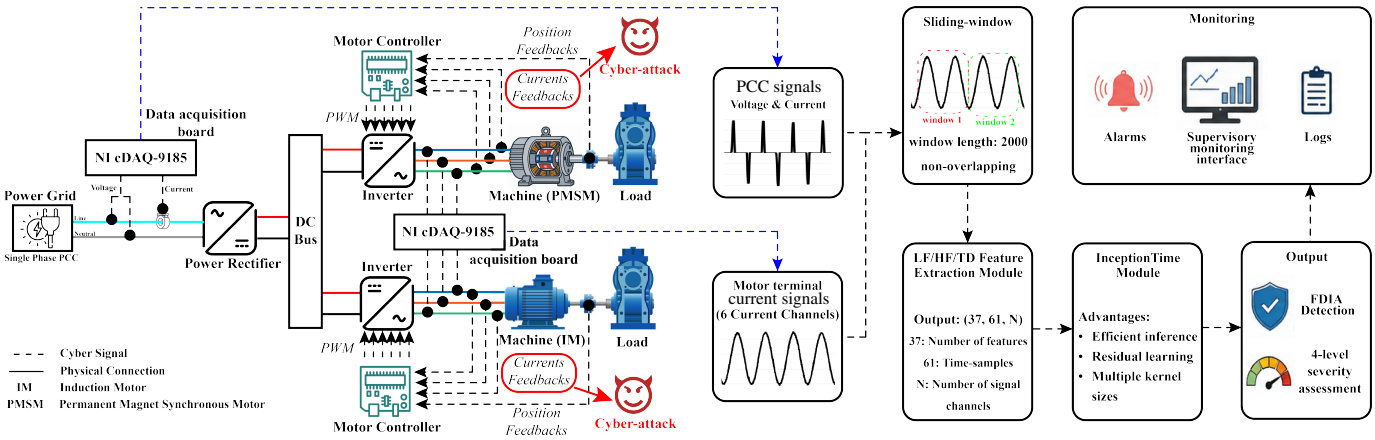


Fig. 1. Experimental platform with a shared PCC, ACIM and PMSM drive branches, monitoring sensors, and current-feedback FDIA injection points.

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Table 1: FDIA dataset structure.

Cases	Phases	Bias Pattern	Total
1–10	A	α_a: 0.0005–0.0050	180
11–20	A,B	$\alpha_a = \alpha_b$: 0.0005–0.0050	180
21–30	A,B	$\alpha_a = 0.0050$, α_b: 0.0005–0.0050	180
		Overall	540

Each case group is evaluated over 3 speeds, 3 load settings, and 2 motor types. For all cases, unspecified phase biases are zero.

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TABLE I
FDIA DATASET STRUCTURE.

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		Overall	540

Each case group is evaluated over 3 speeds, 3 load settings, and 2 motor types. For all cases, unspecified phase biases are zero.

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Table 2: Measurement streams used for monitoring.

Stream	Measured signals	Channels
Motor-side	ACIM and PMSM phase currents	6
PCC-side	PCC voltage and current	2

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TABLE II
MEASUREMENT STREAMS USED FOR MONITORING.

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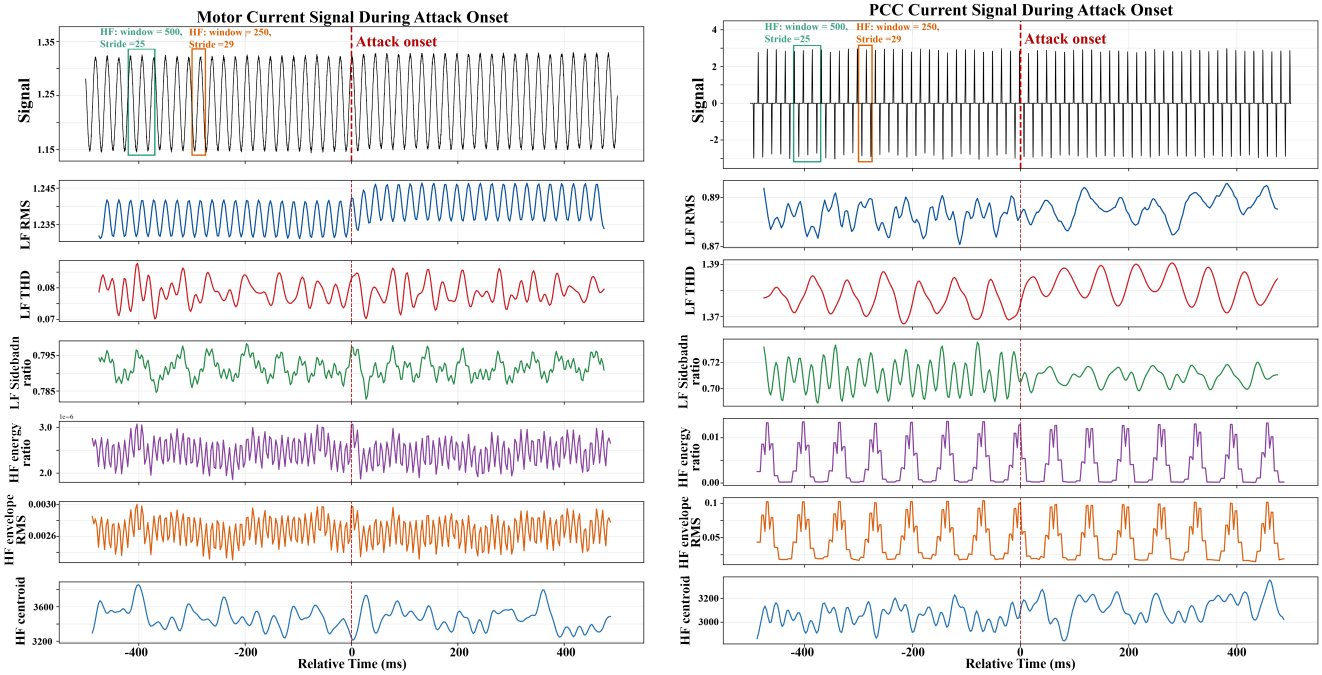


Figure 2: Representative motor-side and PCC-side signals at attack onset with the corresponding LF/HF feature responses.

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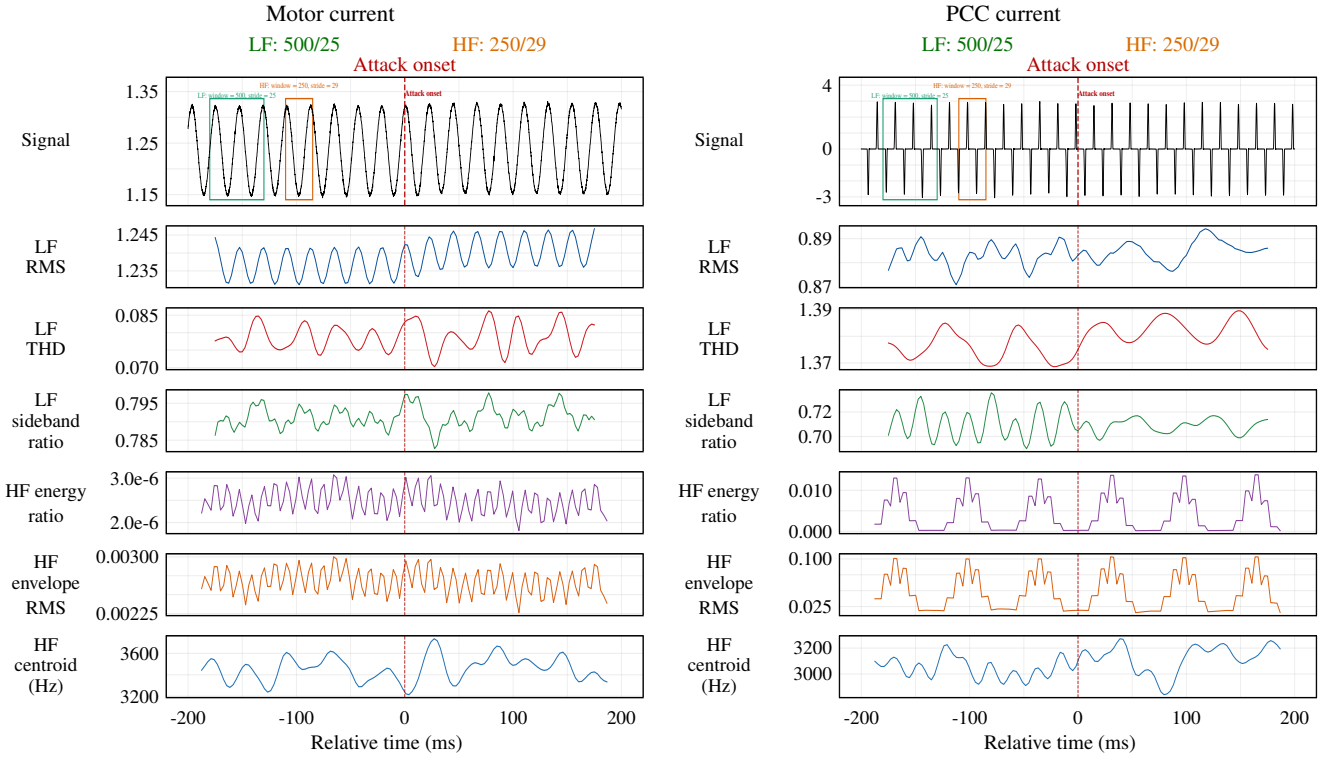


Fig. 2. Representative motor-side and PCC-side signals at attack onset with the corresponding LF/HF feature responses.

distortion, sideband effects, and low-band energy redistribution. The HF branch captures higher-frequency redistribution and envelope-related changes that may appear during small current-feedback FDIAs. The TD features provide complementary sensitivity to amplitude variation, impulsiveness, waveform asymmetry, and short-term signal complexity. Together, these features help make weak PCC FDIA signatures more separable despite harmonics, grid interaction, and aggregation.

B. Inception-Based Detection Model

The extracted LF/HF tensors were modeled using a one-dimensional InceptionTime architecture [15] for multivariate time-series classification. The input dimension is $(61, 222)$ for the motor stream and $(61, 74)$ for the PCC stream. As illustrated in Fig. 3, the network uses bottleneck compression, parallel one-dimensional convolutional kernels of lengths 39, 19, and 9, residual shortcuts, global average pooling, and a dense output layer. Motor and PCC models used stream-specific hyperparameters selected by validation balanced accuracy. The Inception-based design is well suited to the proposed LF/HF representation because the extracted features retain temporal ordering across the 61-step sequence. This preserves temporal interactions among LF, HF, and TD features. Each task uses a separately trained model with two outputs for `fdia2`, three for `fdia3`, and four for `severity4`.

C. Architecture of Online FDIA Detection and Attack-Severity Estimation

The proposed pipeline supports online monitoring from trusted motor-terminal ~~or PCC~~ currents or PCC voltage and current measurements. Figure 4 illustrates the overall architecture of the proposed monitoring pipeline. In the deployed setting, fixed-length windows of ~~current~~ measurement data are acquired from either the motor terminals or the PCC, passed through the LF/HF/TD feature-extraction stage, and then evaluated by the selected learning model to produce anomaly or severity decisions. The measurement, feature-extraction, and inference stages are modular, allowing the same pipeline to operate with motor-side or PCC inputs and supporting feature pruning, reduced-input models, and cross-motor adaptation. The framework provides alarms, severity indicators, and logs for supervisory systems; automatic mitigation is outside the scope. The same architecture is used for binary detection and severity classification~~. Binary detection is the primary PCC deployment task; severity estimates provide secondary diagnostic information~~, with separate normalization and trained models for motor-side and PCC inputs. The experimental results in later sections show that motor-terminal sensing provides cleaner discriminative signatures, whereas PCC sensing offers a less intrusive sensing option for the tested setup.

D. Training Details and Hyperparameters

For the InceptionTime-based models, hyperparameter selection was performed separately for `fdia2` and `severity4` in the motor and PCC streams using validation-set balanced accuracy. The search varied depth ~~, filters, bottleneck size, dropout~~ (3/4/6), filters (16/24/32/48), bottleneck width (8/16/32), dropout (0.20/0.35/0.50), learning rate $(5 \times 10^{-4}/10^{-3})$, weight decay $(5 \times 10^{-5}/10^{-4})$, and batch size (64/128). Table IV summarizes the top-ranked configurations for both sensing streams across all three seeds.

The same 100 candidate configurations were evaluated for two tasks, two sensing streams, and three recording partitions, totaling 1,200 training runs.

Training used Adam and weighted cross-entropy, assigning larger weights to less frequent classes. Each run used up to 60 epochs, with early stopping after 10 epochs without improved validation balanced accuracy. The learning rate was halved after three unimproved epochs, down to 10^{-6}. The best checkpoint and configuration were selected independently within each recording partition using validation balanced accuracy; ties followed the fixed candidate order. Test results were not used for selection. The `fdia3` model retained the `severity4`-selected configuration for each stream and holdout and was trained from fresh weights with three outputs. Its checkpoint was selected by three-class validation balanced accuracy.

Reported test metrics are mean $\pm$ sample standard deviation over the three holdouts. Precision, recall, and F1 are macro-averaged, and balanced accuracy is the mean class recall. The standard deviations describe variation across recording assignments and model initializations, rather than independent physical replications.

E. Domain Adaptation Strategy: Source-Only vs Few-Shot Fine-Tuning

Few-shot adaptation was evaluated by training on ~~one motor type~~ recordings targeting one motor and testing on recordings targeting the other. Two transfer directions were considered: ACIM $\rightarrow$ PMSM and PMSM $\rightarrow$ ACIM. For each direction, the detector was first trained using labeled ~~data from the source motor only~~ source recordings and then evaluated directly on the target ~~motor~~recordings. This *source-only* setting quantifies cross-motor domain shift. Six source recordings were used for training and three for validation, with the 100-candidate search selected by source validation balanced accuracy.

For adaptation, the source-trained model was fine-tuned using a small labeled subset of complete target-domain ~~samples~~ episode groups before testing on ~~held-out target data. The~~ separate target recordings. The nominal target adaptation ratios considered in this work were 1%, 5%, 10%, and 20% of the available target-domain adaptation pool (Table III); the 1% budget selects two of 180 groups (1.11%). Fine-tuning used 20 epochs at 0.2 times the source learning rate, with source normalization fixed. The domain-adaptation strategy was implemented as supervised fine-tuning rather than unsupervised feature alignment. Accordingly, all reported adaptation gains reflect the benefit of limited labeled target-domain data under a practical transfer-learning setting. To ensure reliable comparison, all cross-domain experiments were repeated over three random seeds~~, and the results were reported using both accuracy and balanced accuracy~~.

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Table 3: Dataset dimensions and evaluation protocols.

Parameter	**Value / Configuration**
Data dimensions	
Total samples	60,054
Motor distribution	ACIM: 30,204; PMSM: 29,850
Binary distribution	Normal: 33,054; Attack: 27,000
Input configuration	
Time steps per sample, W	61
Features per channel	37
Channels	Motor: 6; PCC: 2
Motor feature shape	$(N, W, F) = (N, 61, 222)$
PCC feature shape	$(N, W, F) = (N, 61, 74)$
Severity labels	
Raw severity levels	$0\%, 0.05\%, 0.1\%, \ldots, 0.5\%$
Four-class labels	Normal: 33,054; Mild: 8,100; Medium: 10,800; Severe: 8,100
Protocol 1: In-domain evaluation	
Split ratio	Train / validation / test = 70% / 15% / 15%
Sample counts	42,037 / 9,008 / 9,009
Domain composition	ACIM and PMSM samples appear in all splits
Protocol 2: Cross-domain evaluation	
Transfer directions	ACIM $\rightarrow$ PMSM; PMSM $\rightarrow$ ACIM
Adaptation ratios	1%, 5%, 10%, and 20% labeled target data
Reporting	Three seeds; accuracy and balanced accuracy

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TABLE III
DATASET DIMENSIONS AND RECORDING-DISJOINT EVALUATION PROTOCOLS.

Parameter	Value / configuration
Data dimensions	
Complete windows	60,054; 2,000 samples (0.2 s) each
Recordings / episodes	18 recordings; 540 scheduled episodes
Motor distribution	ACIM: 30,204; PMSM: 29,850
Normal / attack windows	33,054 normal; 27,000 attack
Input configuration	
Features per channel	37 (18 LF, 11 HF, 8 TD)
Time steps per sample, W	61
Channels	Motor: 6; PCC: 2
Motor input shape	$(N, 61, 222)$
PCC input shape	$(N, 61, 74)$
Severity labels (`severity4`)	
Normal / low	33,054 / 5,400 windows
Medium / high	7,200 / 14,400 windows
Protocol 1: mixed-motor holdouts	
Recordings per motor	5 training; 2 validation; 2 test
Repeat 1 (seed 1234)	33,718 training; 13,279 validation; 13,057 test windows
Repeat 2 (seed 2234)	33,275 training; 13,250 validation; 13,529 test windows
Repeat 3 (seed 3234)	33,560 training; 13,227 validation; 13,267 test windows
Protocol 2: cross-motor adaptation	
Transfer directions	ACIM $\rightarrow$ PMSM; PMSM $\rightarrow$ ACIM
Target recordings	6 adaptation-pool; 3 held-out test
Target adaptation budgets	1/5/10/20%: 2/9/18/36 complete episode groups, respectively, from 180 groups

Each recording and all its episodes remain in one partition within a repeat. N: number of windows.

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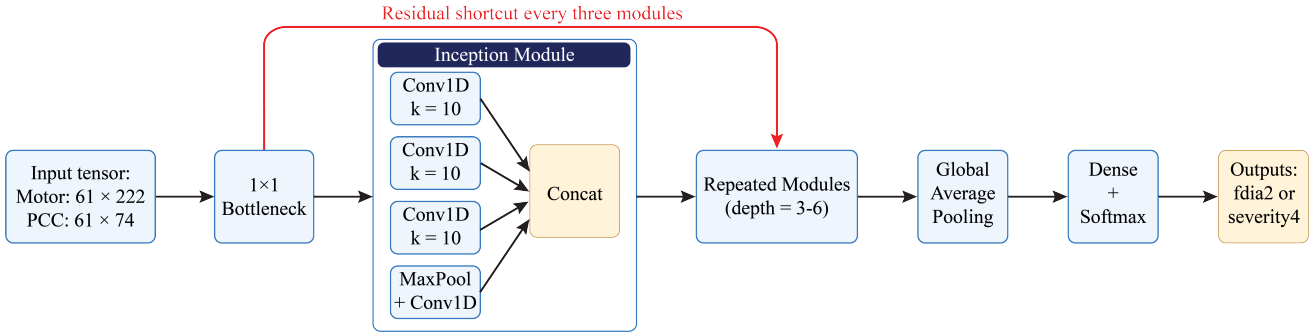


Figure 3: Overview of the InceptionTime-based classifier used for `fdia2` detection and `severity4` classification.

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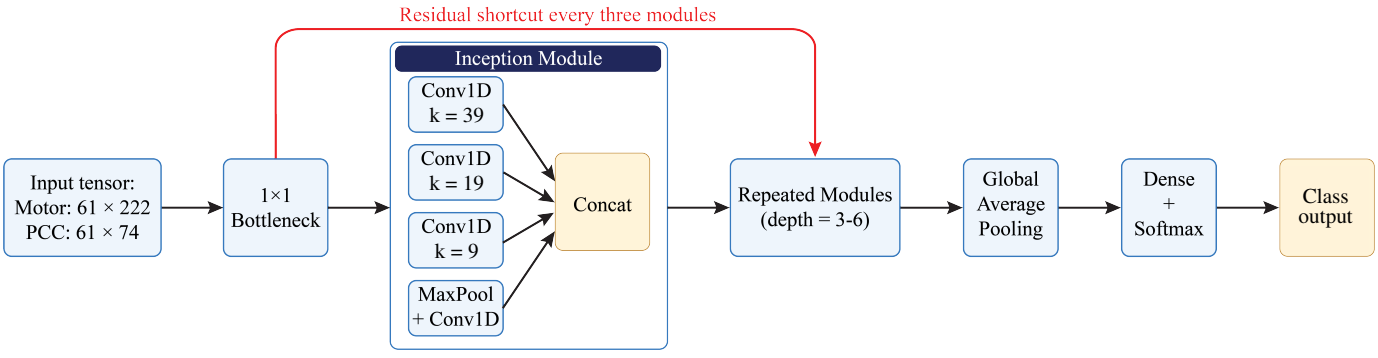


Fig. 3. InceptionTime architecture for FDIA detection, severity assessment, and attack localization.

This protocol is particularly important in the present study because ACIM and PMSM signals exhibit substantial domain differences despite being collected under the same FDIA framework. This tests whether the LF/HF-Inception representation can be adapted with limited target-domain labels.

V. Experimental Results

A. In-Domain Baselines: Motor vs PCC

Table V reports mixed-domain in-domain performance with ACIM and PMSM ~~samples~~ recordings in all splits and no shared recordings or episodes between them. The results show that motor-side sensing provides more discriminative attack signatures than PCC sensing for both ~~(~~`fdia2` ~~) and (~~and `severity4`~~)~~. In addition, ~~(~~`severity4` ~~)~~ is substantially more challenging than binary attack detection, especially at PCC. The `fdia3` rows report joint detection and localization, including the normal class. Fig. 5 shows the `fdia2` and `severity4` confusion matrices, with row percentages pooled over the three holdouts.

Thus, PCC-side severity estimates should be treated as coarse diagnostics, whereas motor-side measurements support finer severity discrimination. Overall, PCC monitoring is more challenging than motor-side monitoring, and severity assessment is harder than binary detection. These baseline findings motivate the subsequent cross-domain analysis, where the additional challenge of motor-type shift is introduced.

Exact-severity recall counts attacks assigned to the correct bias class. Severity-conditioned attack recall also counts attacks assigned to another nonzero bias class. Fig. 6 presents this second view of the PCC severity model: it detects 88.1% of high-bias attacks as some attack level, compared with 70.8% assigned to the exact high-bias class. Thus, an incorrect severity estimate can still provide an attack indication. The same model labels 37.3% of normal windows as attacks, so this detection benefit must be considered together with false alarms. These decisions come from the four-class model; the separately trained binary detector in Fig. 5(b) has fewer false alarms.

B. Comparison with Deep and Classical ML Baselines

To benchmark the proposed LF/HF Inception-based model, representative deep-learning and classical ML baselines were evaluated on the PCC-based `fdia2` and `severity4` tasks, which were the most challenging in-domain settings. The baselines include ResNet, ~~MLSTM-FCN~~LSTM-FCN, Random Forest, ~~and XGBoost,~~ XGBoost, and MultiRocket [16], while the proposed model uses the final stream-specific InceptionTime configuration selected by validation-based hyperparameter search. All methods use the same recording partitions and feature inputs; search budgets differ between model families. As shown in Table VI, the proposed model achieved the highest mean accuracy, balanced accuracy, and F1 score among the evaluated baselines ~~on both PCC tasks. The performance gap is larger for~~ for PCC binary detection, and the highest mean balanced accuracy for PCC severity assessment. For `severity4`~~, confirming that PCC-based severity discrimination is more challenging than binary FDIA detectionand benefits from the proposed temporal LF/HF representation~~, XGBoost achieved higher accuracy and LSTM-FCN slightly higher macro F1. The proposed model therefore improves average class recall for PCC severity assessment, while the preferred model depends on the evaluation metric.

C. Cross-Domain Transfer Without Adaptation

Source-only transfer was evaluated by training on ~~one motor type~~ recordings targeting one motor and testing directly on recordings targeting the other. Two transfer directions were considered: ACIM → PMSM and PMSM → ACIM. Table VII shows that source-only ~~fdia2~~ `fdia2` transfer falls near chance balanced accuracy for ~~both sensingstreams~~ACIM-to-PMSM transfer, while the reverse direction falls below chance for motor sensing and reaches 56.00% for PCC sensing. For `fdia2`, the source-only results in Table VII show a substantial degradation relative to the mixed-domain baselines. This confirms a substantial ACIM/PMSM domain shift in the tested transfer setting. Thus, reliable cross-motor FDIA detection in this setting benefits from target-domain adaptation.

D. Domain Adaptation Results: 1/5/10/20%

Table VIII reports ~~fdia2~~ `fdia2` few-shot adaptation using nominal budgets of 1%, 5%, 10%, and 20% labeled target ~~data. In all cases~~episode groups. At the 20% budget, adaptation improved performance relative to the source-only setting in all cases, although the magnitude of the gain was strongly direction dependent. For ACIM-to-PMSM transfer, adaptation ~~gives only modest gains, reaching 57.49~~reaches 75.99% balanced accuracy for motor sensing and ~~60.06~~56.30% for PCC at 20% target data. For PMSM-to-ACIM transfer, adaptation is much stronger, reaching ~~86.10~~92.63% balanced accuracy for motor sensing and ~~91.16~~91.38% for PCC at 20% target data. With only the nominal 1% budget, PMSM-to-ACIM balanced accuracy rises from 36.08% to 64.35% for motor sensing and from 56.00% to 77.43% for PCC sensing, gains of 28.27 and 21.43 percentage points.

Overall, the adaptation results show that few-shot target-domain fine-tuning can improve transfer performance, but the gains are strongly direction dependent and are much larger in the PMSM → ACIM direction. This asymmetry indicates that cross-motor FDIA adaptation cannot be treated as a symmetric transfer problem; in practice, the pretraining motor domain materially affects PCC deployment performance.

E. Error Analysis: Confusion Matrices and FP/FN Behavior

Fig. 5 compares the in-domain confusion matrices for `fdia2` and `severity4`. For `fdia2`, motor-side sensing yields a strongly diagonal confusion matrix ~~with low~~ for attack windows, with pooled false-positive and false-negative rates of ~~0.75% and 1.56~~20.9% and 2.3%, respectively~~, whereas PCC sensing increases these rates to 6.55% and 15.46% because the attack is observed only through the aggregated current response~~. PCC sensing yields 14.7% false positives and 24.6% false negatives, showing fewer false alarms but more

ORIGINAL — Table 4 (main.tex)

Table 4: Top five InceptionTime configurations ranked by validation balanced accuracy on `severity4`.

Rank	D	F	BN	Dr	LR	WD	BS	test BA (%)
Motor stream								
1	3	32	16	0.20	0.0010	5×10^{-5}	128	98.25
2	3	32	8	0.20	0.0010	10^{-4}	64	98.08
3	3	16	16	0.35	0.0010	5×10^{-5}	128	98.02
4	3	32	8	0.20	0.0010	5×10^{-5}	128	97.99
5	3	32	8	0.35	0.0010	10^{-4}	64	97.96
PCC stream								
1	6	32	32	0.20	0.0010	10^{-4}	128	80.22
2	6	48	8	0.20	0.0005	5×10^{-5}	128	79.95
3	3	48	32	0.35	0.0005	5×10^{-5}	64	79.84
4	4	32	32	0.35	0.0005	10^{-4}	64	79.56
5	4	48	32	0.50	0.0010	10^{-4}	128	79.40

D: depth; F: filters; BN: bottleneck; Dr: dropout; LR: learning rate; WD: weight decay; BS: batch size; test BA: test balanced accuracy.

REVISED — Table 4 (main_revised.tex)

TABLE IV

TOP FIVE `SEVERITY4` CONFIGURATIONS BY MEAN VALIDATION BALANCED ACCURACY OVER THREE SEEDS (%).

Rank	D	F	BN	Dr	LR	WD	BS	Seed 1234	Seed 2234	Seed 3234	Mean ± SD
MOTOR stream											
1	6	48	32	0.35	0.001	5×10^{-5}	64	90.62	93.12	95.52	93.09 ± 2.45
2	6	24	16	0.20	0.001	10^{-4}	128	91.53	93.47	92.77	92.59 ± 0.99
3	3	16	16	0.50	0.001	5×10^{-5}	64	89.68	92.91	93.98	92.19 ± 2.24
4	3	24	16	0.50	0.001	10^{-4}	128	89.26	91.96	95.18	92.13 ± 2.97
5	3	16	16	0.50	0.001	10^{-4}	64	90.85	92.44	92.76	92.02 ± 1.02
PCC stream											
1	6	32	32	0.50	0.001	10^{-4}	64	52.25	57.15	58.55	55.98 ± 3.31
2	4	48	32	0.20	0.001	5×10^{-5}	64	51.10	58.28	58.10	55.82 ± 4.09
3	6	48	32	0.35	0.001	5×10^{-5}	64	51.94	56.00	58.19	55.38 ± 3.17
4	6	24	16	0.50	0.001	5×10^{-5}	128	50.09	56.16	59.68	55.31 ± 4.86
5	4	32	32	0.50	0.001	5×10^{-5}	64	50.10	57.31	58.16	55.19 ± 4.42

D: depth; F: filters; BN: bottleneck; Dr: dropout; LR: learning rate; WD: weight decay; BS: batch size. SD is the sample standard deviation. This table summarizes validation performance across repeats; models for test evaluation are selected independently within each seed, not by this aggregate ranking.

ORIGINAL — Table 5 (main.tex)

Table 5: In-domain performance under the mixed ACIM/PMSM split.

Task	Accuracy (%)	Bal. Acc. Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
PCC stream					
`fdia2`	89.44	88.99	89.71	88.99	89.25
`severity4`	84.17	80.22	80.50	80.22	80.32
Motor type	100.00	100.00	100.00	100.00	100.00
Motor stream					
`fdia2`	98.89	98.85	98.91	98.85	98.88
`severity4`	98.68	98.25	98.22	98.25	98.24
Motor type	100.00	100.00	100.00	100.00	100.00

Precision, recall, and F1 are reported as macro-averaged values in paper.

REVISED — Table 5 (main_revised.tex)

TABLE V

IN-DOMAIN PERFORMANCE ON HELD-OUT ACIM/PMSM RECORDINGS (%).

Task	Acc.	BA	Prec.	Recall	F1
PCC stream					
`fdia2`	80.78 (±2.95)	80.29 (±3.03)	81.04 (±3.14)	80.29 (±3.03)	80.42 (±2.96)
`severity4`	61.00 (±1.32)	55.62 (±2.94)	52.93 (±2.36)	55.62 (±2.94)	52.96 (±2.15)
`fdia3`	74.32 (±4.16)	77.84 (±4.63)	76.16 (±4.56)	77.84 (±4.63)	75.52 (±4.14)
MOTOR stream					
`fdia2`	87.41 (±11.13)	88.26 (±9.99)	89.90 (±7.84)	88.26 (±9.99)	87.23 (±11.39)
`severity4`	87.64 (±8.87)	90.81 (±4.29)	86.09 (±9.44)	90.81 (±4.29)	86.06 (±8.95)
`fdia3`	85.64 (±9.60)	89.55 (±4.47)	87.95 (±8.29)	89.55 (±4.47)	86.70 (±8.18)

Cells show mean above (± sample SD), over three holdouts; this is not a range or confidence interval. BA: balanced accuracy. Precision, recall, and F1 are macro-averaged.

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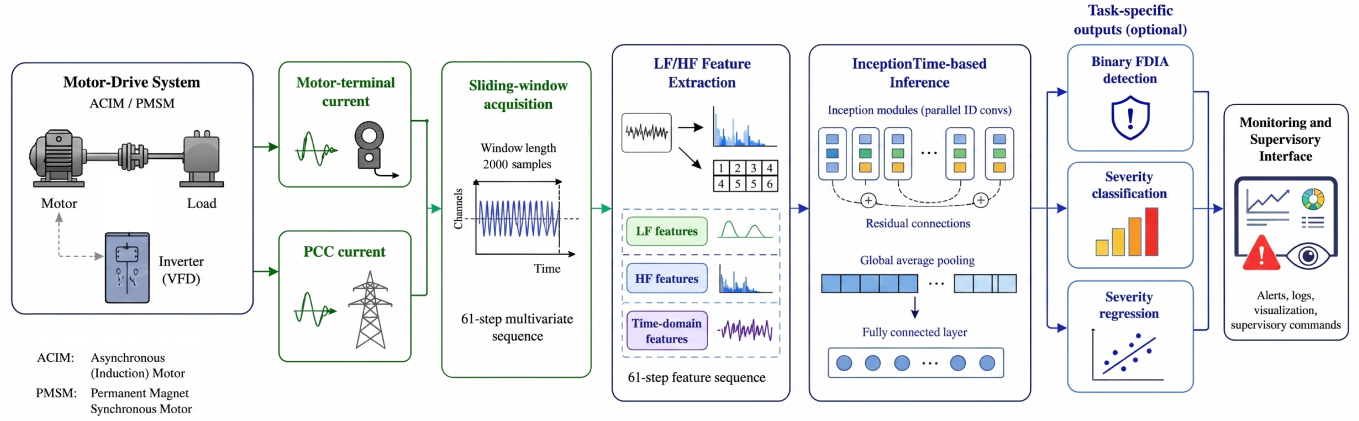


Figure 4: Online FDIA detection and severity-assessment architecture for motor-side and PCC-side sensing.

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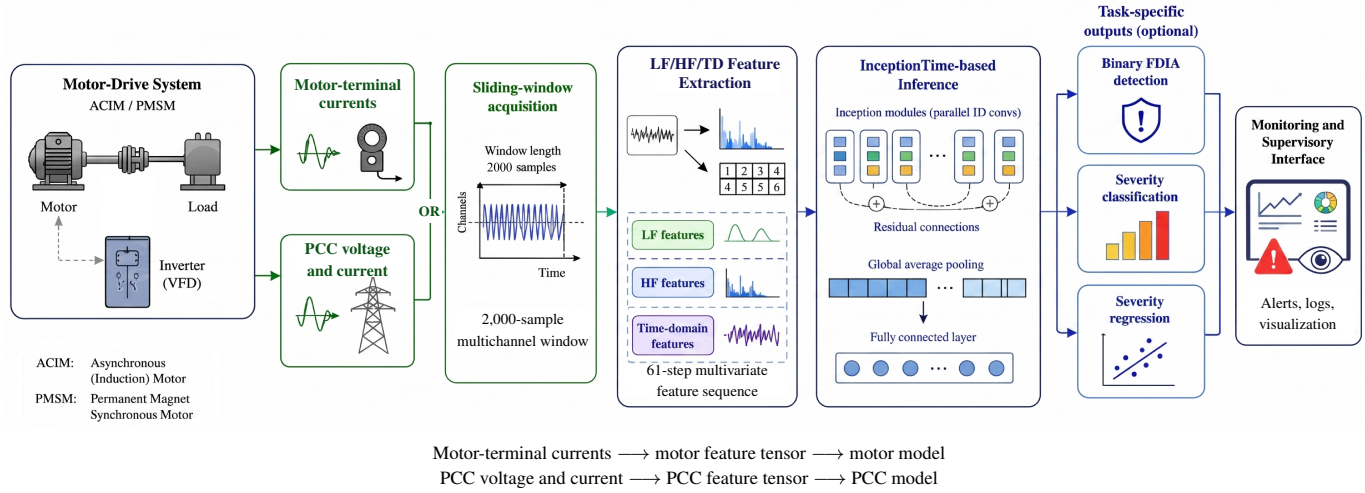


Fig. 4. Online FDIA detection and severity-assessment architecture using separate motor-terminal and PCC sensing configurations.

ORIGINAL — Table 6 (main.tex)

Table 6: Baseline comparison on the PCC-based `fdia2` and `severity4` tasks.

Method	Acc. (%)	Balanced Acc. (%)	F1 (%)
fdia2			
Proposed InceptionTime	89.44	88.99	89.25
ResNet	85.58	84.83	85.20
MLSTM-FCN	84.58	83.80	84.17
Rand. Forest	86.05	85.27	85.67
XGBoost	86.45	85.75	86.11
severity4			
Proposed InceptionTime	84.17	80.22	80.32
ResNet	77.31	70.22	70.68
MLSTM-FCN	77.96	71.11	71.89
Rand. Forest	83.42	72.68	77.40
XGBoost	79.00	65.21	70.45

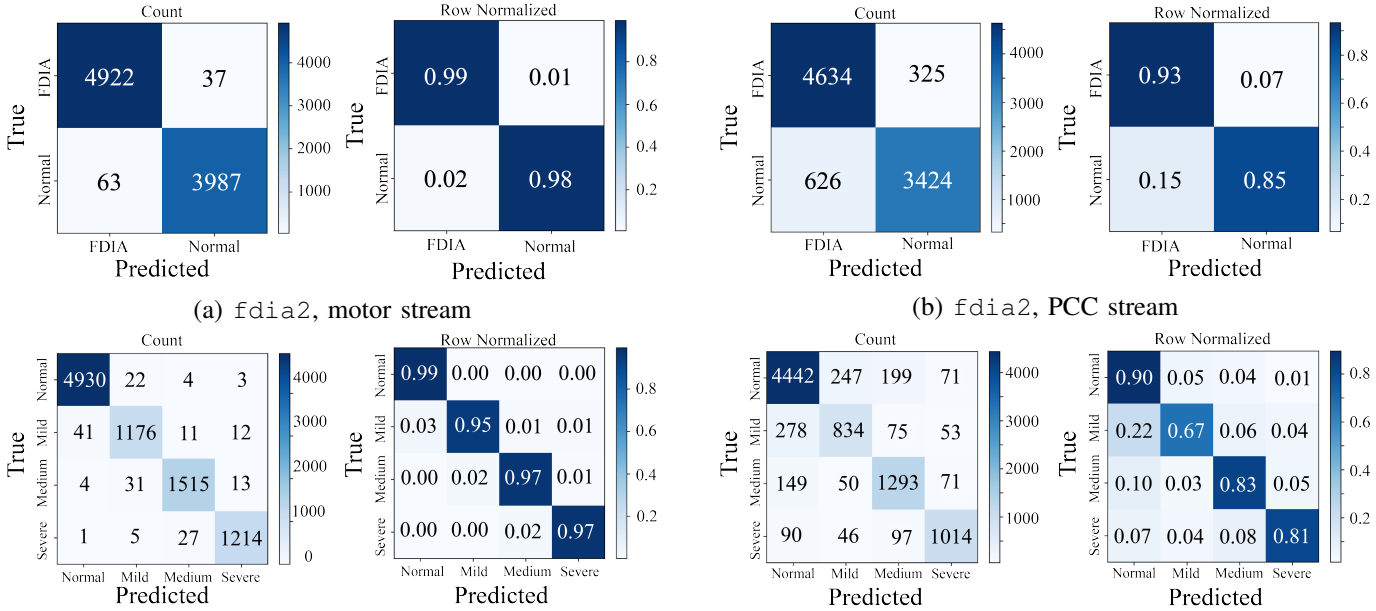
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TABLE VI
PCC BASELINE COMPARISON ON `FDIA2` AND `SEVERITY4` (%).

Method	Accuracy	BA	Macro F1
fdia2			
InceptionTime	**80.78** ± 2.95	**80.29** ± 3.03	**80.42** ± 2.96
ResNet	73.91 ± 5.78	73.99 ± 5.33	73.76 ± 5.66
LSTM-FCN	72.57 ± 4.89	72.52 ± 4.31	72.35 ± 4.66
Rand. Forest	70.43 ± 1.00	68.69 ± 0.74	68.45 ± 0.85
XGBoost	75.37 ± 0.81	73.84 ± 1.11	73.97 ± 1.17
MultiRocket	70.10 ± 2.13	69.55 ± 2.82	69.35 ± 2.39
severity4			
InceptionTime	61.00 ± 1.32	**55.62** ± 2.94	52.96 ± 2.15
ResNet	65.18 ± 5.35	50.41 ± 5.44	51.15 ± 5.71
LSTM-FCN	66.60 ± 1.72	52.38 ± 2.12	**53.23** ± 1.94
Rand. Forest	67.36 ± 0.17	43.49 ± 2.18	45.07 ± 2.59
XGBoost	**70.45** ± 0.74	47.17 ± 2.02	49.67 ± 2.17
MultiRocket	58.41 ± 3.34	44.92 ± 3.46	44.28 ± 2.52

Mean ± sample SD over three holdouts. Bold: highest mean for each metric within a task; it does not indicate statistical significance.

ORIGINAL — Figure 5 (main.tex)



(a) `fdia2`, motor stream (b) `fdia2`, PCC stream

(c) `severity4`, motor stream (d) `severity4`, PCC stream

Figure 5: In-domain confusion matrices for the proposed LF/HF InceptionTime classifier: (a) `fdia2` with motor-side measurements, (b) `fdia2` with PCC measurements, (c) `severity4` with motor-side measurements, and (d) `severity4` with PCC measurements.

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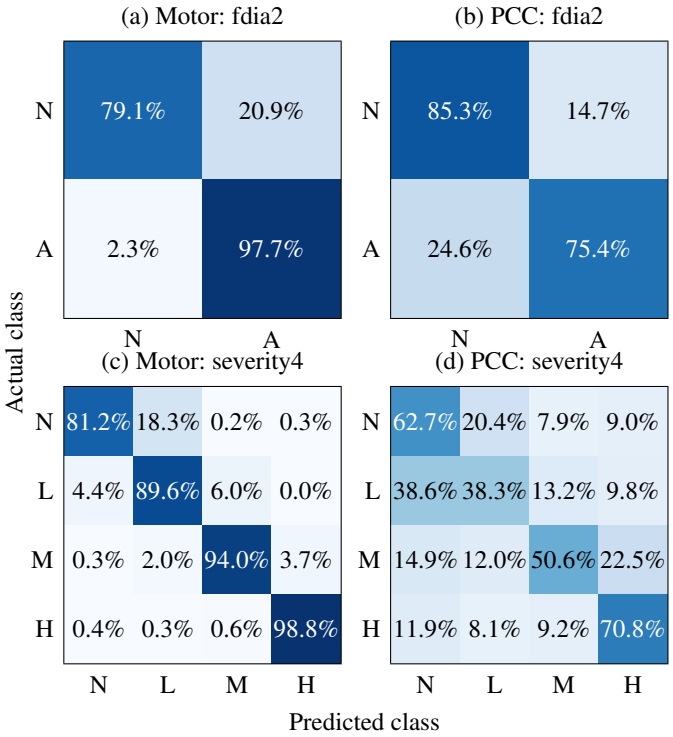


Fig. 5. In-domain confusion matrices (row percentages). N/A: normal/attack; N/L/M/H: normal/low/medium/high bias.

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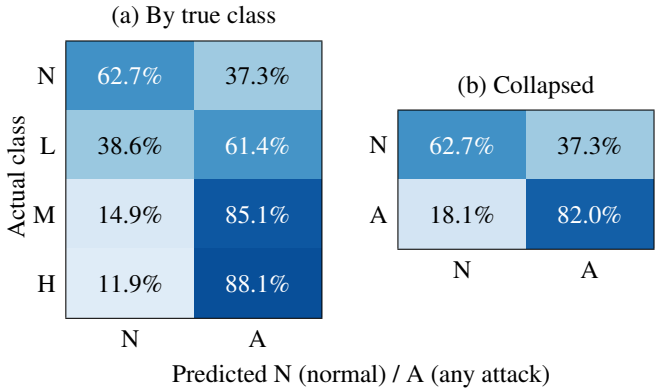


Fig. 6. PCC severity-model decisions: (a) attack recall by true severity; (b) collapsed normal/attack matrix (row percentages).

ORIGINAL — Table 7 (main.tex)

Table 7: Cross-domain source-only performance without target-domain adaptation on `fdia2` task.

Stream	Direction	Accuracy (%)	Balanced Acc. (%)
Motor	ACIM → PMSM	53.87	50.10
Motor	PMSM → ACIM	51.08	51.80
PCC	ACIM → PMSM	54.79	50.07
PCC	PMSM → ACIM	52.38	57.03

REVISED — Table 7 (main_revised.tex)

TABLE VII
SOURCE-ONLY FDIA2 TRANSFER WITHOUT TARGET ADAPTATION (%).

Direction	Accuracy	BA	Macro F1
MOTOR stream			
ACIM → PMSM	55.21 ± 1.14	50.60 ± 0.73	37.71 ± 2.37
PMSM → ACIM	36.22 ± 4.23	36.08 ± 4.11	33.89 ± 4.62
PCC stream			
ACIM → PMSM	55.88 ± 0.64	51.58 ± 1.52	41.13 ± 5.84
PMSM → ACIM	51.94 ± 6.54	56.00 ± 5.69	43.62 ± 11.38

Mean ± sample SD over three holdouts. Directions are different target-domain tests, so no best direction is highlighted.

ORIGINAL — Table 8 (main.tex)

Table 8: Few-shot domain-adaptation results for `fdia2` using labeled target-domain fine-tuning. Values are mean test balanced accuracy (%) over three seeds.

Stream	Direction	Source-only	1%	5%	10%	20%
Motor	ACIM → PMSM	50.10	50.06	49.63	53.15	57.49
Motor	PMSM → ACIM	51.80	57.95	73.75	78.86	86.10
PCC	ACIM → PMSM	50.07	51.79	55.51	57.47	60.06
PCC	PMSM → ACIM	57.03	70.10	84.56	89.12	91.16

REVISED — Table 8 (main_revised.tex)

TABLE VIII
FEW-SHOT FDIA2 ADAPTATION BY LABELED TARGET-DOMAIN FINE-TUNING: BALANCED ACCURACY (%).

Direction	Source only	1%	5%	10%	20%
MOTOR stream					
ACIM → PMSM	50.60 ± 0.73	50.43 ± 1.74	53.83 ± 1.14	57.51 ± 7.24	$\mathbf{75.99} \pm 1.73$
PMSM → ACIM	36.08 ± 4.11	64.35 ± 9.23	73.93 ± 8.78	88.04 ± 5.36	$\mathbf{92.63} \pm 3.72$
PCC stream					
ACIM → PMSM	51.58 ± 1.52	54.01 ± 0.12	54.62 ± 1.76	54.04 ± 1.35	$\mathbf{56.30} \pm 3.23$
PMSM → ACIM	56.00 ± 5.69	77.43 ± 10.43	84.22 ± 4.62	86.72 ± 4.37	$\mathbf{91.38} \pm 2.06$

Mean ± sample SD over three holdouts. Bold: highest observed mean within each direction and stream; no test-based configuration selection. Budgets refer to the target episode-group pool.

missed attacks. For severity4, PCC errors ~~concentrate between adjacent classes, especially~~ include both normal/~~mild and mild/medium~~attack confusion and confusion among the nonzero bias classes, indicating that PCC measurements retain coarse attack information but blur finer severity distinctions.

Fig. 7 complements the aggregate confusion-matrix results by showing a representative held-out ACIM detection sequence on a continuous experimental signal. ~~Both sensing locations follow the~~ The motor-side detector follows most scheduled normal/FDIA intervals, ~~but the motor-side detector produces cleaner transitions than~~ while the PCC-side detector exhibits missed attacks and false alarms, especially near the first ~~attack onset and low-severity FDIA intervals~~low-bias attack. These results indicate that motor-side measurements are more separable, while PCC measurements remain useful but less reliable for ~~fine severity discrimination~~detecting small attacks. RMS traces use 500-sample frames at stride 25, and predictions are placed at window ends. Green curves show attack probability, red curves show predicted labels, and dashed gray curves show true labels; the dotted threshold is 0.5.

VI. LATENCY AND REDUCED-FEATURE INFERENCE ANALYSIS

Practical PCC monitoring requires low latency and stable online throughput. This section reports per-window latency, throughput, pruning, and reduced-feature effects. ~~In particular, the analysis focuses on end-to-end processing latency, online throughput, feature-pruning effects, and practical considerations for deployment at the PCC.~~

A. Inference Latency and Throughput on CPU/GPU

Runtime was benchmarked on an Intel Core Ultra 9 285K CPU and NVIDIA RTX 5090 GPU using the selected seed-1234 LF/HF-Inception binary pipelines. The benchmark was split into three stages: (i) LF/HF feature extraction from one 2,000-sample raw window, (ii) normalization, host-to-device transfer, and model inference on the resulting feature tensor, and (iii) end-to-end per-window execution. ~~GPU inference was measured with and without host-to-device transfer~~Each variant used 10 warm-up runs and 50 timed repetitions, with randomized variant order. Processing starts from an available measurement window; acquisition and disk I/O are excluded.

Table IX summarizes the measured per-window latencies. For the PCC stream, LF/HF feature extraction required ~~21.72~~22.30 ms on CPU~~and the Inception model needs only 0.71 ms on GPU. When host-to-device transferwas included, the GPU inference stage remained sub-millisecond at 0.74~~, while normalization, transfer, and GPU inference required 0.78 ms. As a result, the median PCC end-to-end ~~latency was only 22.46 msfor End-to-End inference~~processing latency was 23.03 ms. A similar pattern was observed for the motor stream. Feature extraction required ~~65.25~~66.38 ms on CPU, while ~~model inference required 0.36 ms on GPU. Including transfer, the GPU inference stage was 0.38~~normalization, transfer, and GPU inference required 1.42 ms, leading to ~~an~~ a median end-to-end latency of ~~65.63~~67.81 ms. ~~For the tested PCC pipeline, the measured end-to-end latency was 22.46 ms under the reported hardware configuration.~~ In terms of throughput, the end-to-end PCC pipeline ~~sustained 44.53~~ had a reciprocal-median processing rate of 43.42 windows/s, while the corresponding motor pipeline sustained ~~15.24~~ 14.75 windows/s.

B. Feature Pruning for Low-Latency Anomaly Detection

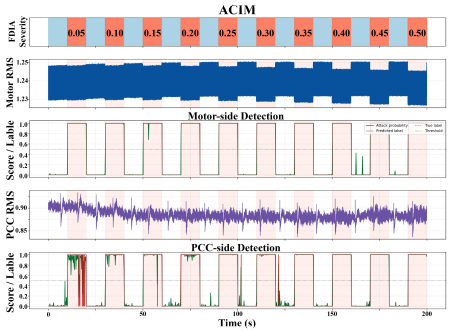
Feature pruning was evaluated for the fdia2 task to reduce online latency. Features were ranked by permutation importance, defined as the validation balanced-accuracy drop caused by permuting each feature ~~column~~trajectory across windows, averaged over three permutations. Test labels were not used for ranking. Reduced subsets were then formed using global top-75%, top-50%, and top-25% rankings, as well as per-channel top-15 and top-10 selections. Because the benchmark uses selective LF/HF feature computation, only retained features ~~are extracted~~and their required intermediate quantities are calculated; therefore, the measured extraction latency reflects the computational benefit of pruning. Each reduced subset was retrained using the full model's selected architecture and training protocol.

Table X shows that pruning can reduce latency while preserving, and in some cases improving, detection performance. For the PCC stream, ~~the best tradeoff~~ a compact representation was obtained with the top-25% subset, which retained 18 of 74 features. This subset ~~improved balanced accuracy from 88.99% to 90.11~~retained the full model's mean balanced accuracy of 80.29% and reduced end-to-end CPU+GPU latency from ~~25.38~~23.03 ms to ~~19.71~~12.44 ms, corresponding to ~~a 1.29×~~ an 85.18% speedup. For the motor stream, the top-25% subset retained 56 of 222 features, ~~improved~~ reduced balanced accuracy from ~~98.85% to 99.48~~88.26% to 86.83%, and reduced latency from ~~109.07~~67.81 ms to ~~56.21~~27.87 ms, corresponding to a ~~1.94×~~ 143.33% speedup.

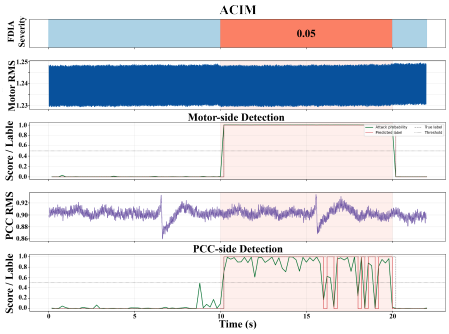
~~The intermediate subsets did not always reduce runtime, even when they contained fewer features~~All evaluated subsets reduced runtime, but the reduction was not proportional to feature count. This indicates that pruning effectiveness depends not only on feature count, but also on which LF, HF, and TD ~~feature families remain active~~calculations remain necessary. The feature-importance rankings in Fig. 8 further show that the two sensing streams rely on different features: motor-side detection is dominated by TD and selected LF features, whereas PCC-side detection places greater emphasis on harmonic and waveform-shape features. Bars show mean validation importance and whiskers show one standard deviation across three holdouts. Motor channels 1–3/4–6 are ACIM/PMSM currents; PCC channels 1/2 are voltage/current.

Overall, the LF/HF representation contains redundancy for anomaly detection, and structured pruning can improve both latency and predictive performance. This is especially relevant for PCC-based monitoring, where the compact top-25% subset provides ~~both the fastest and most accurate PCC detector~~the lowest measured PCC processing time while preserving mean balanced accuracy.

ORIGINAL — Figure 6 (main.tex)



(a) 200 s overview



(b) 0–22 s zoomed view

Figure 6: Representative continuous-signal FDIA detection example for the ACIM. Panel (a) shows the 200-s sequence, and panel (b) zooms into the minor attack transition.

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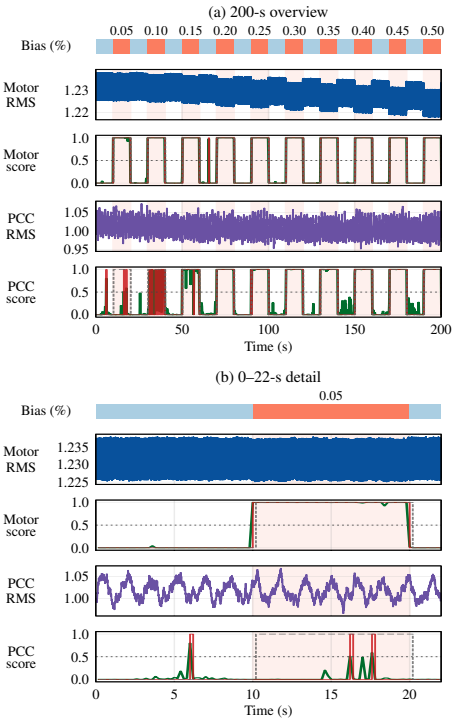


Fig. 7. Held-out ACIM detection: (a) 200-s overview and (b) 22-s detail of the first attack onset. Shading denotes attack intervals.

ORIGINAL — Table 9 (main.tex)

Table 9: Measured per-window latency and throughput of the GPU-assisted LF/HF-InceptionTime pipeline.

Stream	Feature Extraction (ms)	GPU Inference (ms)	GPU Transfer + Inference (ms)	End-to-End (ms)	Throughput (windows/s)
PCC	21.72	0.71	0.74	22.46	44.53
Motor	65.25	0.36	0.38	65.63	15.24

REVISED — Table 9 (main_revised.tex)

TABLE IX
GPU-ASSISTED PER-WINDOW LATENCY AND PROCESSING RATE.

Statistic	Feature extract. (ms)	Norm./transfer + inference (ms)	Total processing (ms)	Rate (windows/s)
PCC stream				
Median	22.30	0.78	23.03	43.42
95th pct.	24.15	0.95	25.03	–
MOTOR stream				
Median	66.38	1.42	67.81	14.75
95th pct.	75.52	1.58	76.98	–

Rate is the reciprocal of median total processing time. Stages are timed within each iteration; marginal percentiles need not sum. Acquisition and disk I/O are excluded.

ORIGINAL — Figure 7 (main.tex)

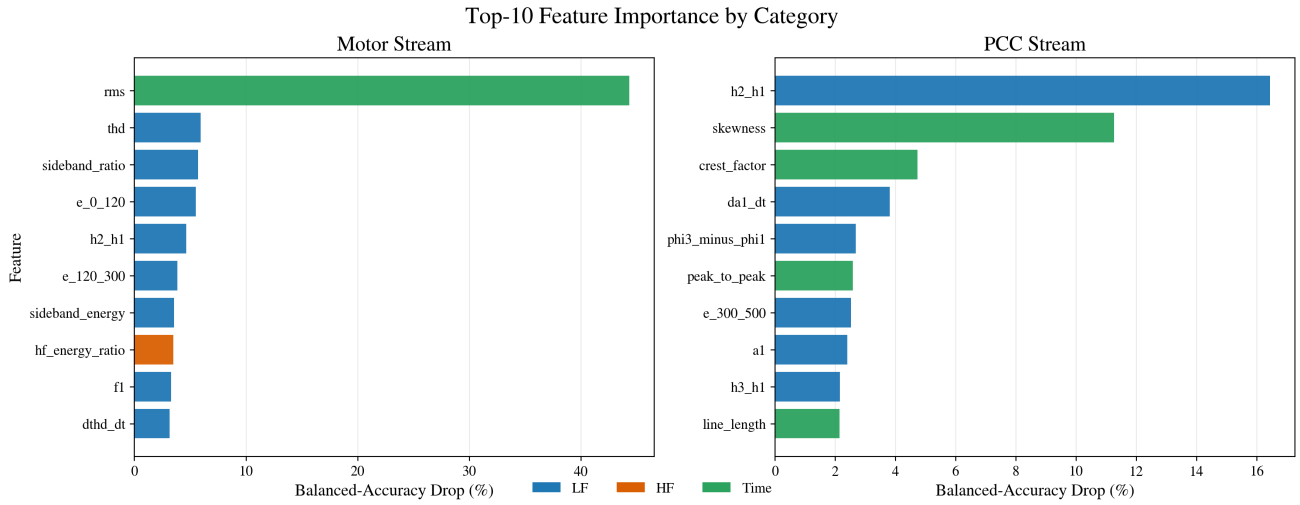


Figure 7: Feature-importance rankings for the motor and PCC streams.

REVISED — Figure 8 (main_revised.tex)

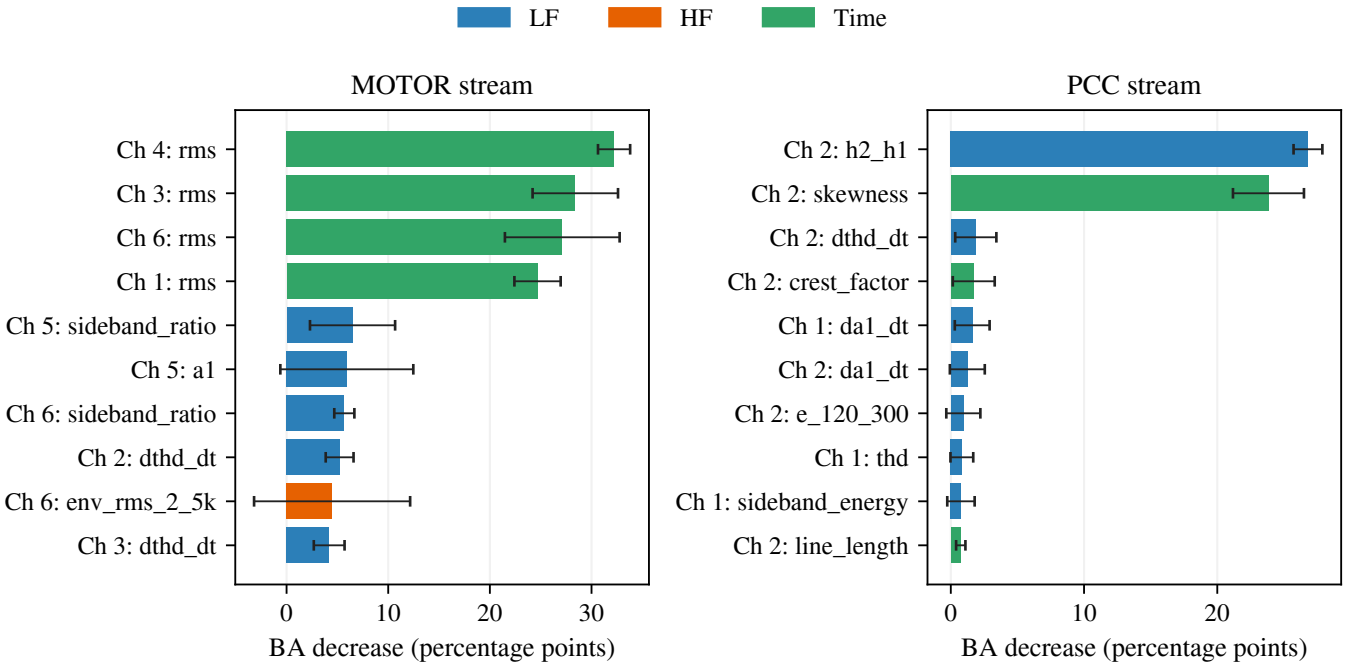


Fig. 8. Top ten features by validation permutation importance for motor and PCC detection. Error bars show one standard deviation.

C. LF/HF/TD Ablation for Binary Anomaly Detection

A complementary LF/HF/TD ablation was performed on the `fdia2` task to identify which feature groups are most useful for binary anomaly detection in the PCC and motor streams. All seven combinations were retrained using the selected full-feature architecture within each stream and split. Detailed ablation results are reported in Table XI. For the PCC stream, the LF+Time subset achieved ~~88.67~~80.03% balanced accuracy, close to the full model at ~~88.99~~80.29%, whereas HF-only and HF+Time dropped to ~~73.98% and 79.78~~64.49% and 67.85%, respectively. For the motor stream, the time-only subset ~~matched or slightly~~ exceeded the full model, reaching ~~99.45~~92.89% balanced accuracy compared with ~~98.85~~88.26% for the full representation, while HF-only dropped to ~~65.34~~63.76% and HF+Time reached 94.11%. Overall, the ablation confirms that PCC detection relies mainly on LF+Time information, motor-side detection is driven primarily by TD features, and HF features ~~are the least informative group in both streams~~alone are less informative but complement TD features for motor sensing. This trend is consistent with the pruning results and supports reduced-feature inference in the tested pipeline. These ablations assess feature contributions and were not used to select a replacement primary model.

VII. Discussion

A. Why Adaptation Helps More in One Direction

The adaptation results show that ACIM and PMSM do not form a symmetric transfer pair. PMSM → ACIM adaptation is much more effective than ACIM → PMSM adaptation, particularly for PCC sensing. In the PCC stream, ~~the ACIM → PMSM model tends to collapse toward the normal class, whereas the~~ Table VIII shows a substantial PMSM → ACIM ~~model retains more attack-sensitive structure and can be corrected with limited labeled ACIM data. Thus, the source motor affects the initial class bias, the retained target-relevant information, and the benefit obtained from~~ gain even at the nominal 1% budget, while the reverse direction remains less effective. These results show that the benefit of few-shot fine-tuning depends on the source and target motor types; the physical causes of the asymmetry were not isolated. For practical PCC deployment, a small amount of target-domain calibration data is therefore important when ~~the deployed motor family differs~~ monitoring attacks on a motor family absent from the source ~~training domain~~attack examples.

B. PCC vs Motor Tradeoffs

The results reveal a consistent tradeoff between sensing convenience and predictive separability. Motor-side measurements provide the clearest attack-sensitive signatures across nearly all tasks. In the in-domain experiments, the motor stream outperformed the PCC stream for both `fdia2` and `severity4`, and the confusion matrices showed substantially stronger diagonal structure for motor-side classification. ~~The motor stream also achieved lower regression error in the attack-only severity-regression task, indicating that direct motor-current sensing preserves richer information about attack intensity and progression.~~

PCC remains the more deployment-oriented option because it avoids direct terminal access and matches shared industrial power infrastructure. A shared PCC sensing point also avoids dedicated instrumentation at each motor, offering a path toward scalable monitoring; larger installations remain to be validated. Despite lower separability, PCC achieved useful in-domain detection and strong adapted performance, especially for PMSM → ACIM transfer.

Overall, the motor stream is the stronger sensing modality when maximum classification performance is the priority, especially for fine-grained severity discrimination. The PCC stream, by contrast, is the more deployment-oriented choice because it avoids invasive sensing while still supporting accurate and computationally efficient FDIA detection. The value of the framework is therefore not in surpassing motor-side sensing, but in improving the accuracy and computational efficiency of PCC-based monitoring on the tested platform.

C. Limitations

Several limitations should be noted when interpreting the present results. First, the study considers only two motor families, ACIM and PMSM, under ~~a fixed experimental FDIA framework~~additive current-feedback bias on one experimental platform. The recordings use fixed speed/load settings; dynamic changes and converter topologies were not varied. The three recording holdouts reuse this apparatus and are not independent hardware replications. Localization was evaluated with one attacked branch at a time; simultaneous attacks on both motors were not tested. Because the attacked branch is fixed within each recording, operating-condition cues may also contribute to localization. Broader validation across additional machines, loads, control modes, sensing setups, and attack families is needed before generalizing beyond the tested platform.

Second, the domain-adaptation analysis is based on supervised few-shot fine-tuning with labeled target-domain data. This does not cover cases where target labels are unavailable or delayed. Unsupervised or self-supervised adaptation strategies may therefore be required for more constrained deployment scenarios.

Third, the deployment analysis was performed using a GPU-assisted software implementation and focused mainly on the binary FDIA detection task; embedded implementation and task-specific optimization for severity assessment remain future work.

Finally, ~~some baselines were competitive under macro-F1; therefore, the conclusions are strongest for accuracy and~~ PCC severity classification and false alarms remain limitations. Some severity baselines achieved higher accuracy or macro F1; the proposed model's severity advantage is in mean balanced accuracy, not universal model dominance.

VIII. Conclusion

This paper experimentally evaluated PCC-based detection of low-severity current-feedback FDIAs in a heterogeneous ACIM/PMSM inverter-fed drive platform. Motor-terminal measurements provided the strongest separability, but PCC

measurements supported useful nonintrusive detection despite aggregation, harmonics, and grid-side interactions. In the mixed-domain setting with separate recordings and episodes, the proposed LF/HF-InceptionTime method achieved ~~strong~~ useful `fdia2` detection from both sensing locations, while PCC `severity4` estimates were less reliable and should be interpreted as coarse diagnostics. Cross-motor experiments showed substantial ACIM/PMSM domain shift, making source-only transfer unreliable. Few-shot target-domain fine-tuning improved transfer performance, with much larger gains for PMSM $\rightarrow$ ACIM adaptation than for ACIM $\rightarrow$ PMSM adaptation. A nominal 1% target-episode budget improved PMSM-to-ACIM PCC balanced accuracy by 21.43 percentage points. Runtime and pruning results further indicate that the PCC pipeline is compatible with the required online processing rate. Overall, the results indicate that PCC-based FDIA monitoring is promising on the tested ACIM/PMSM platform under additive current-feedback attacks.

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Appendix A
Additional Experimental and Diagnostic Results

This appendix provides ~~supplementary material supporting the main results, including representative FDIA episodes and the extracted LF/HF/TD feature set~~representative FDIA recordings, the extracted feature definitions, severity-conditioned detection metrics, hyperparameter sensitivity, and auxiliary bias regression. Complete numeric results and per-repeat configurations accompany the manuscript.

A. Representative FDIA Episodes

Figure 9 shows representative consecutive FDIA episodes applied to the ACIM and PMSM drive systems. In both examples, the top panel shows the scheduled FDIA severity level, while the middle and bottom panels show the corresponding motor-current and PCC-current RMS responses during the first 200 ~~s of operation~~s of each selected recording. RMS uses 500-sample frames at stride 25, matching the time-domain RMS definition. Complete-recording diagnostics retain the post-schedule normal tails.

B. Extracted Feature Set

Table XII summarizes the extracted features used in the proposed LF/HF/TD pipeline. Each channel is represented using low-frequency spectral features, high-frequency spectral-envelope features, and time-domain statistical features.

C. Severity-Conditioned Attack Detection

Table XIII separates correct bias-bin classification from detecting any attack. For confusion matrix C, severity-conditioned attack recall is

$$R_{\text{attack}|k} = \frac{\sum_{j=1}^{3} C_{kj}}{\sum_{j=0}^{3} C_{kj}}, \qquad k \in \{1,2,3\}, \tag{3}$$

while exact-severity recall is $C_{kk}/\sum_j C_{kj}$. The repeated false-alarm entries belong to the same four-class model and do not depend on the actual attack level. Fig. 6 presents the corresponding PCC decision matrices; pooled normal recall can differ slightly from the mean per-repeat recall because normal-window counts vary across holdouts.

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Table 10: Effect of feature pruning on `fdia2` performance and runtime.

Feature Set	No. Feature	Balanced Accuracy (%)	F1 (%)	Feature Extraction (%)	End-to-End (ms)	Speedup (×)	Runtime Gain(%)
PCC stream							
Full	74	88.99	89.25	24.61	25.38	1.00	0
75%	56	88.65	88.88	22.11	22.91	1.11	11
50%	37	88.90	89.04	22.26	23.00	1.10	10
25%	18	90.11	90.28	18.88	19.71	1.29	29
15/ch.	30	89.33	89.52	33.95	34.70	0.73	-27
10/ch.	20	89.40	89.61	19.09	20.23	1.25	25
Motor stream							
Full	222	98.85	98.88	108.65	109.07	1.00	0
75%	166	99.24	99.27	110.15	110.74	0.98	-2
50%	111	99.39	99.38	112.59	113.20	0.96	-4
25%	56	99.48	99.51	55.84	56.21	1.94	94
15/ch.	90	99.49	99.50	104.02	104.60	1.04	4
10/ch.	60	99.46	99.47	62.19	62.79	1.74	74

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TABLE X
FEATURE PRUNING FOR `FDIA2`: PERFORMANCE AND RUNTIME.

Subset	No.	BA (%)	F1 (%)	Extract. (ms)	Total (ms)	Speedup (%)
PCC stream						
Full	74	80.29 ± 3.03	80.42 ± 2.96	22.30	23.03	0.00
75%	56	78.28 ± 3.07	78.41 ± 3.37	19.65	20.33	13.26
50%	37	78.87 ± 2.79	79.10 ± 2.97	16.63	17.31	33.07
25%	18	80.29 ± 2.14	80.58 ± 2.29	11.81	**12.44**	85.18
15/ch.	30	77.59 ± 4.75	77.74 ± 4.94	15.64	16.25	41.76
10/ch.	20	**80.47** ± 2.79	**80.69** ± 2.80	12.07	12.68	81.60
MOTOR stream						
Full	222	88.26 ± 9.99	87.23 ± 11.39	66.38	67.81	0.00
75%	166	87.95 ± 11.38	87.19 ± 12.42	56.28	57.70	17.52
50%	111	89.28 ± 11.33	88.20 ± 12.87	45.50	46.89	44.62
25%	56	86.83 ± 10.93	85.44 ± 12.39	26.57	**27.87**	143.33
15/ch.	90	**92.10** ± 9.71	**91.44** ± 10.88	37.68	38.96	74.04
10/ch.	60	88.18 ± 9.28	87.20 ± 10.62	28.48	29.80	127.56

Performance: mean ± sample SD. Timing: seed-1234 GPU-assisted median; 95th percentiles are supplied in the results. Bold: highest BA/F1 or lowest total latency within a stream. Only selected statistics and their required intermediate calculations are evaluated. Percentage speedup is $100(T_{\text{full}}/T_{\text{subset}} - 1)$, using median total processing times.

ORIGINAL — Table 11 (main.tex)

Table 11: LF/HF/TD ablation results for `fdia2`.

Feature Group	Selected Features	Accuracy (%)	Balanced Accuracy (%)	F1 (%)
PCC stream				
Full	74	89.44	88.99	89.25
LF only	36	88.31	87.89	88.11
HF only	22	74.43	73.98	74.06
Time only	16	80.48	79.71	79.99
LF + HF	58	84.04	83.66	83.80
LF + Time	52	89.22	88.67	88.99
HF + Time	38	80.19	79.78	79.89
Motor stream				
Full	222	98.89	98.85	98.88
LF only	108	89.77	89.48	89.62
HF only	66	66.06	65.34	65.41
Time only	48	99.47	99.45	99.46
LF + HF	174	88.98	88.63	88.81
LF + Time	156	99.03	99.00	99.02
HF + Time	114	99.07	99.00	99.06

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TABLE XI
LF/HF/TD ABLATION FOR `FDIA2` (%).

Features	No.	Accuracy	BA	Macro F1
PCC stream				
Full	74	**80.78** ± 2.95	**80.29** ± 3.03	**80.42** ± 2.96
LF only	36	74.03 ± 3.84	73.56 ± 3.13	73.59 ± 3.33
HF only	22	64.50 ± 4.62	64.49 ± 4.56	64.21 ± 4.71
Time only	16	72.18 ± 1.73	70.66 ± 1.77	70.65 ± 1.76
LF + HF	58	68.66 ± 4.60	68.15 ± 4.26	68.18 ± 4.39
LF + Time	52	80.49 ± 1.73	80.03 ± 1.79	80.16 ± 1.73
HF + Time	38	69.49 ± 3.66	67.85 ± 3.11	67.72 ± 3.06
MOTOR stream				
Full	222	87.41 ± 11.13	88.26 ± 9.99	87.23 ± 11.39
LF only	108	81.61 ± 3.08	80.54 ± 2.55	80.94 ± 2.77
HF only	66	63.32 ± 4.96	63.76 ± 4.28	63.14 ± 4.81
Time only	48	92.61 ± 5.51	92.89 ± 4.96	92.58 ± 5.50
LF + HF	174	80.33 ± 7.88	79.91 ± 7.06	80.05 ± 7.60
LF + Time	156	90.70 ± 8.32	91.17 ± 7.45	90.65 ± 8.35
HF + Time	114	**93.87** ± 4.56	**94.11** ± 4.04	**93.85** ± 4.54

Mean ± sample SD over three holdouts. Bold: highest mean within each stream; these test comparisons do not select a replacement primary model.

ORIGINAL — Table 12 (main.tex)

Table 12: Extracted features per channel in the proposed LF/HF/TD pipeline.

Type	No.	Window / Stride	Features
LF	18	500 / 25	A_1, f_1, ϕ_1, THD, H_2/H_1, H_3/H_1, H_5/H_1, H_7/H_1, $\phi_3 - \phi_1$, $\phi_5 - \phi_1$, sideband energy/ratio, E_{0-120}, $E_{120-300}$, $E_{300-500}$, df_1/dt, dA_1/dt, $d\text{THD}/dt$
HF	11	500 / 25	RMS_{2-5k}, energy ratio, centroid, bandwidth, flatness, kurtosis, peak frequency, peak-to-band, envelope RMS, envelope kurtosis, envelope crest factor
TD	8	500 / 25	RMS, standard deviation, crest factor, kurtosis, skewness, peak-to-peak, line length, Teager MAV
Total	37	–	LF (18), HF (11), TD (8)

REVISED — Table 12 (main_revised.tex)

TABLE XII
EXTRACTED FEATURES PER CHANNEL IN THE PROPOSED LF/HF/TD PIPELINE.

Type	No.	Window / Stride	Features
LF	18	500 / 25	A_1, f_1, ϕ_1, THD, H_2/H_1, H_3/H_1, H_5/H_1, H_7/H_1, $\phi_3 - \phi_1$, $\phi_5 - \phi_1$, sideband energy/ratio, E_{0-120}, $E_{120-300}$, $E_{300-500}$, df_1/dt, dA_1/dt, $d\text{THD}/dt$
HF	11	250 / 29	RMS_{2-5k}, energy ratio, centroid, bandwidth, flatness, kurtosis, peak frequency, peak-to-band, envelope RMS, envelope kurtosis, envelope crest factor
TD	8	500 / 25	RMS, standard deviation, crest factor, kurtosis, skewness, peak-to-peak, line length, Teager MAV
Total	37	–	LF (18), HF (11), TD (8)

D. *Hyperparameter Sensitivity*

One-factor sensitivity experiments changed depth to 3/9, dropout to 0.1/0.4, or learning rate to $3 \times 10^{-4}/3 \times 10^{-3}$ around each selected configuration, retaining all other settings. Fig. 10 reports these perturbations. PCC binary mean balanced accuracy ranges from 72.23% to 78.33% among these variants, compared with 80.29% for the selected reference. PCC severity varies from 51.25% to 55.71%, compared with 55.62%. Motor severity is more sensitive to the high learning rate, falling to 77.69% mean balanced accuracy. The PCC severity limitation persists across the evaluated settings, although classification performance remains sensitive to the training parameters.

E. *Auxiliary Diagnostic Tasks*

The joint detection and localization task (`fdia3`) assigns 33,054 windows to normal operation, 13,500 to ACIM attack, and 13,500 to PMSM attack. The model uses a three-class softmax output and the same recording-separated holdouts as the other tasks. Its results are included in Table V. Mean normal false-alarm rates are 22.85% for motor sensing and 33.12% for PCC sensing.

For attack-only regression, the target is $100\|\boldsymbol{\alpha}\|_\infty$, in percentage points of nominal current-feedback bias. A scalar output replaces the classification head; training uses Smooth-L1 loss with transition parameter 0.5 and validation mean absolute error (MAE) for checkpoint selection. The severity-selected architecture is retained within each stream and repeat. Motor-side MAE is 0.0196 ± 0.0041 percentage points, with mean $R^2 = 0.961$; PCC MAE is 0.0981 ± 0.0049 percentage points, with mean $R^2 = 0.267$. Fig. 11 shows prediction distributions at every tested bias level for each repeat. These are estimates of injected amplitude on attacked windows, not measured mechanical impact.

ORIGINAL — Figure 8 (main.tex)

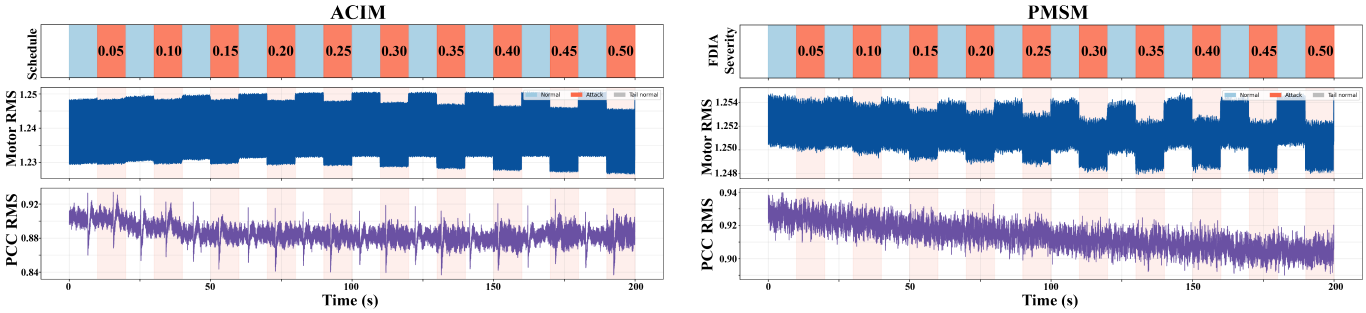


Figure 8: Representative consecutive FDIA episodes applied to the ACIM and PMSM motor-drive systems. Top: scheduled FDIA severity; middle: motor-current RMS; bottom: PCC-current RMS.

REVISED — Figure 9 (main_revised.tex)

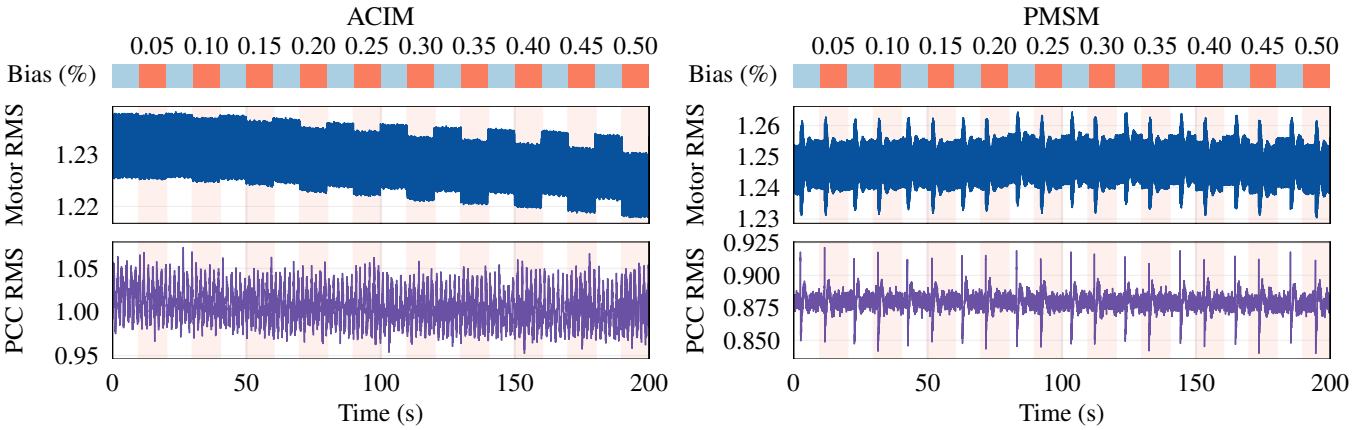


Fig. 9. Representative ACIM/PMSM attack episodes: injected bias (top), motor-current RMS (middle), and PCC-current RMS (bottom), in acquisition units.

REVISED — Table 13 (main_revised.tex)

TABLE XIII
EXACT-SEVERITY AND ANY-ATTACK RECALL FROM FOUR-CLASS PREDICTIONS (%).

True bias	Exact recall	Any-attack recall	Normal false alarms
MOTOR stream			
Low	89.58 ± 2.14	95.58 ± 4.15	19.08 ± 16.08
Medium	93.96 ± 1.05	99.71 ± 0.04	19.08 ± 16.08
High	98.76 ± 0.27	99.65 ± 0.29	19.08 ± 16.08
PCC stream			
Low	38.33 ± 7.04	61.39 ± 2.64	37.23 ± 3.17
Medium	50.58 ± 7.01	85.08 ± 6.52	37.23 ± 3.17
High	70.79 ± 2.28	88.09 ± 7.22	37.23 ± 3.17

Mean $\pm$ sample SD over three holdouts. False alarms refer to the same normal class for all three attack levels. Different metrics are not competing methods, so none is highlighted as best.

REVISED — Figure 10 (main_revised.tex)

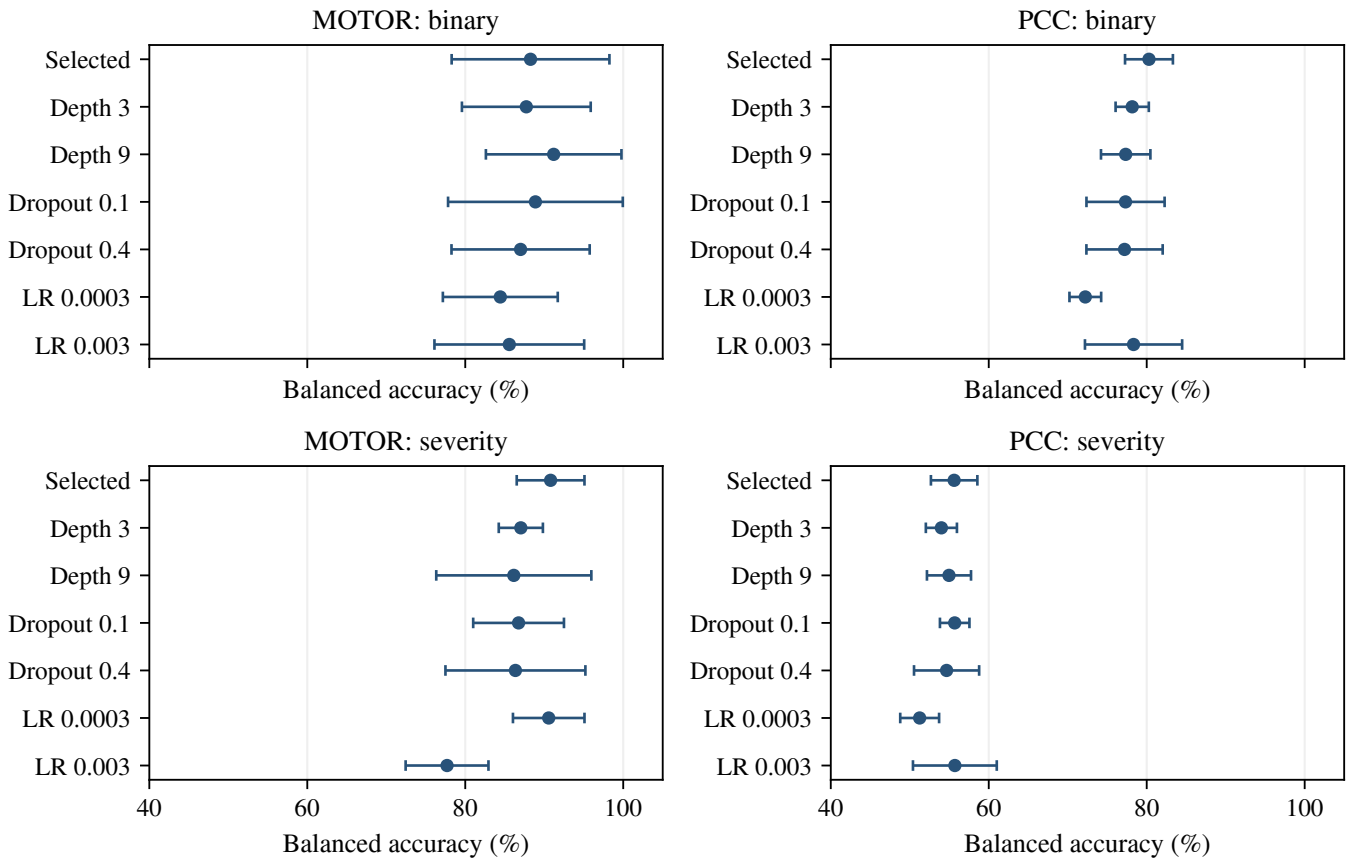


Fig. 10. One-factor sensitivity around the validation-selected full-feature configuration in each repeat. Error bars indicate sample SD across three recording holdouts. Only the named hyperparameter changes; the selected reference can differ across tasks, streams, and repeats.

REVISED — Figure 11 (main_revised.tex)

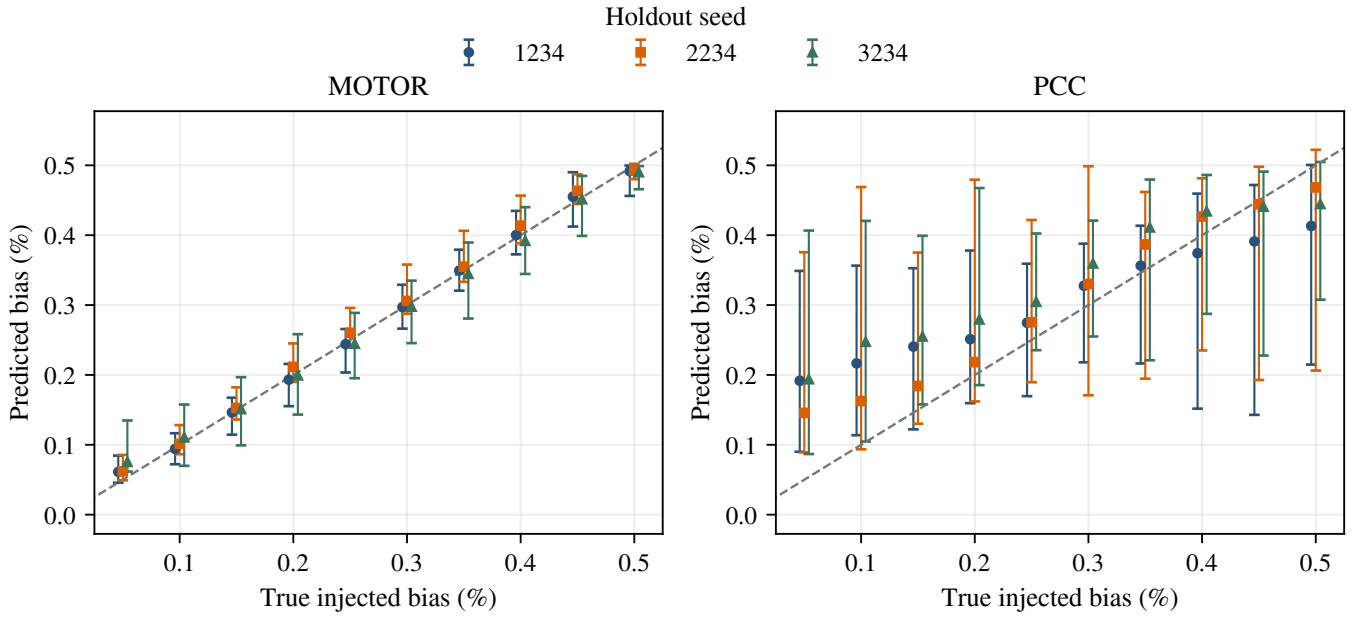


Fig. 11. Attack-only injected-bias regression on held-out recordings. Points and whiskers show the median and 10th–90th percentiles of predictions at each true bias, separately for each holdout seed; slight horizontal offsets separate repeats. The dashed diagonal is ideal agreement. All held-out attack windows are included; repeated appearances across holdouts are not independent recordings.